

A Course in Modern Macroeconomics

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Complete Solutions

Chapters 1–4 (23 exercises)

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These four chapters cover **Aula 1** (Kurlat ch. 1–2) and **Aula 2** (Kurlat ch. 3 and §4.1–4.2), plus §4.3–4.5 which belong to **Aula 3**. Together they are the source material for **Lista de Exercícios 1**.

Ch.	Title	Exercises	Pages
1	GDP	1.1–1.5	27–29
2	Beyond GDP	2.1–2.10	40–44
3	Basic Facts about Economic Growth	3.1–3.3	51–52
4	The Solow Growth Model	4.1–4.5	72–74
Total		23	

On verification — please read

Kurlat provides no answer key. The book has no “Answers to Odd-Numbered Problems” section and no published solutions manual; it ends with the bibliography (pp. 318–319). Every result below was therefore verified by other means, and the method used is stated explicitly at each step:

- **Executed numerical check** — marked ✓. Every number was computed in Python and the code is reproduced so you can re-run it.
- **Internal consistency** — the three GDP methods are made to agree; steady states are substituted back; limiting and special cases are checked.
- **Cross-book** — for canonical models (Solow, factor prices) the method is checked against Romer (2012) ch. 1 and its solutions manual, both in Livros/. Notation differs; conclusions must not.

Where an exercise requires downloading data (2.1, 3.1–3.3), no numerical answer is asserted. Those are marked as *data exercises*: the method, the expected qualitative pattern and runnable code are given instead.

Contents

1	Chapter 1	GDP	2
1.1	GDP Accounting — the Three Methods		2
1.2	Real GDP, Index Numbers and Chain Weighting		6
1.3	The Boundaries of GDP		12
2	Chapter 2	Beyond GDP	13
2.1	The HDI and the Jones–Klenow Welfare Measure		14
2.2	Valuing Life Expectancy and the Composition of Consumption		19
2.3	Risk Aversion, Inequality and the Veil of Ignorance		27
3	Chapter 3	Basic Facts about Economic Growth	33
3.1	Long-Run Growth and Historical Comparisons		33
3.2	The Kaldor Facts		35
3.3	Growth Across Countries and Convergence		37
4	Chapter 4	The Solow Growth Model	40
4.1	Comparative Statics — Shocks to Capital and Labour		40
4.2	Factor Markets and Factor Prices		45
4.3	Alternative Production Technologies		48
	Appendix — Cross-reference to the course		56

Chapter 1 GDP

The three methods. GDP is the market value of final goods and services produced domestically in a period. It can be measured three equivalent ways:

- **Production:** sum of *value added* = output – intermediate inputs.
- **Expenditure:** $Y = C + I + G + (X - M)$.
- **Income:** wages + profits + interest + rents + indirect taxes – subsidies.

The three must give the same number; forcing them to agree is the best check on any accounting answer.

Traps. Intermediate goods are netted out (no double counting). Inventory changes are investment. Imports are subtracted because they appear inside C , I or G but were not produced domestically. Transfers (taxes, pensions) are *not* production. Government output is valued at input cost.

Real vs. nominal. Real GDP holds prices fixed. Using base-year prices (a Laspeyres-type index) and current-year prices (Paasche-type) give *different* growth rates whenever relative prices move — the substitution bias. The **chain-weighted** measure takes the geometric (Fisher) average of adjacent-year links, which is why national accounts use it.

Across countries. Converting at market exchange rates understates poor countries' output because non-tradables are cheap there. **PPP** values every country's basket at a common set of prices.

GDP Accounting — the Three Methods

Problem 1.1 Accounting

Question

How does GDP accounting record the following events? For each of them, describe how they would be computed in GDP accounts using the income method, the production method and the expenditure method.

- (a) A car manufacturer buys components from Japan for \$1 to be used in production later on and stores them at its warehouse.
- (b) A car manufacturer buys components from Japan for \$1 and uses half of those components in the production of a car, that it sells to Andy for \$2. It stores the rest of the components.
- (c) An army battalion is deployed to the border to repel a threatened Canadian invasion. The soldiers earn wages of \$10,000 and use ammunition that the government buys for \$5,000. The ammunition is produced using \$2,000 of imported steel and 100 hours of work, for which the workers were paid \$1,000.
- (d) Walmart sells 1000 bottles of Coca-Cola for \$1,500. It had previously paid \$1,200 for them.
- (e) A shipyard builds a cruise ship. It pays wages of \$200,000, interest on loans (from US residents) of \$100,000 and \$300,000 for imported raw materials. The ship is sold for \$1,000,000 to a cruise company. In the same year, the cruise company has revenue for \$50,000 from operating cruises, pays wages of \$20,000 to its workers and has no other expenses. Half the cruise revenue comes from tourists who reside in the United States and half comes from tourists who reside abroad.
- (f) The government collects \$1000 in income taxes from Roger.
- (g) Roger earns \$4000 for working as a babysitter and pays \$1000 in income taxes.

Solution:

Strategy. For each event compute all three ways and require them to agree. That agreement is the verification — there is no answer key to check against.

- (a) **Contribution to GDP: \$0.**

Method	Entries	Total
Production	no domestic output; imported components are not US production	\$0
Expenditure	I (inventories) = +\$1; $M = \$1 \Rightarrow -\1	\$0
Income	no wages, profits or interest generated domestically	\$0

The purchase raises *inventory investment* by \$1 and imports by \$1. The two cancel exactly. This is the general principle: buying a foreign good never changes GDP at the moment of purchase, whatever the buyer does with it.

- (b) **Contribution to GDP: \$1.50.**

Of the \$1 of components, \$0.50 is used up in production and \$0.50 is stored.

Method	Entries	Total
Production	output \$2 – intermediate inputs used \$0.50	\$1.50
Expenditure	$C = \$2$; $I = +\$0.50$ (unused components); $M = \$1$	\$1.50
Income	profits = $2 - 0.50 = \$1.50$ (no wages stated)	\$1.50

✓ $2 + 0.50 - 1 = 1.50$. Note that only the *half actually consumed in production* is netted out as an intermediate input. The stored half is still an asset of the firm, so it sits in inventories.

(c) **Contribution to GDP: \$13,000.**

Two producers here: the government (which produces defence services) and the ammunition manufacturer.

Method	Entries	Total
Production	government value added = \$10,000 (valued at input cost) ammunition maker = $5,000 - 2,000 = \$3,000$	\$13,000
Expenditure	$G = 10,000 + 5,000 = \$15,000$; $M = \$2,000$	\$13,000
Income	soldiers' wages \$10,000 + ammo workers' wages \$1,000 + ammo firm profits ($5,000 - 2,000 - 1,000$) = \$2,000	\$13,000

✓ Three methods agree. Two points worth stating explicitly:

- **Government output is valued at input cost** because there is no market price for repelling an invasion. So the soldiers' \$10,000 of wages *is* the measured value of the defence service produced.
- The ammunition is an **intermediate input** into defence from the production side, but from the expenditure side it is a *final* purchase by government. Both routes give \$13,000 — they just split the same total differently.

(d) **Contribution to GDP this year: \$300.**

Method	Entries	Total
Production	retail value added = $1,500 - 1,200 = \$300$	\$300
Expenditure	$C = \$1,500$; I (inventories) = $-\$1,200$	\$300
Income	Walmart's wages + profits = \$300	\$300

Walmart produces a *retail service* — getting the bottles onto a shelf near the customer — and that service is worth \$300. The \$1,200 of Coca-Cola was counted as GDP when it was *produced*; selling it now is a drawdown of inventories, which is exactly why the expenditure side subtracts \$1,200.

(e) **Contribution to GDP: \$750,000.**

Shipyard. Value added = $1,000,000 - 300,000 = \$700,000$. Income check: wages 200,000 + interest 100,000 + profits ($1,000,000 - 300,000 - 200,000 - 100,000$) = 400,000, total \$700,000. ✓

Cruise company. No intermediate inputs, so value added = revenue = \$50,000.
Income: wages 20,000 + profits 30,000 = \$50,000. ✓

Method	Entries	Total
Production	700,000 + 50,000	\$750,000
Expenditure	$I = \$1,000,000$ (the ship is a capital good) $C = \$25,000$ (US tourists); $X = \$25,000$ (foreign tourists) $M = \$300,000$	\$750,000
Income	700,000 + 50,000	\$750,000

✓ $1,000,000 + 25,000 + 25,000 - 300,000 = 750,000$.

Three things to notice: the ship is **investment**, not consumption, because it is a durable asset bought by a firm to produce with. **Interest paid to US residents is domestic income**, so it stays inside value added — it is a distribution of the \$700,000, not a cost that reduces it. And cruises sold to foreign residents are **exports** even though the service is consumed inside the country.

(f) **Contribution to GDP: \$0.**

A tax is a **transfer**: purchasing power moves from Roger to the government, but nothing is produced. All three methods record zero. In the income method in particular, taxes are not a new source of income — income is measured *before* the transfer, and counting it again at the government would double count.

(g) **Contribution to GDP: \$4,000.**

Method	Entries	Total
Production	babysitting services produced, no intermediate inputs	\$4,000
Expenditure	$C = \$4,000$ by the households who hired him	\$4,000
Income	Roger's labour income, <i>pre-tax</i>	\$4,000

The \$1,000 of tax does not reduce GDP — by (f) it is a transfer. Contrast this with the *same* babysitting done by a parent for their own child, which is real production but is **excluded** from GDP because no market transaction records it. That boundary problem is the subject of Problem 1.4.

Every line above obeys one rule: **GDP counts production, and only production, once.** Imports are subtracted because they were produced elsewhere; intermediates are netted out because they are already inside the final good; transfers are ignored because they move claims, not output.

Real GDP, Index Numbers and Chain Weighting

Problem 1.2 Comparisons Across Time and Across Countries

Question

Suppose these are the prices (in US dollars) and quantities of goods A and B produced in the US in 2017 and 2018:

	p_A	q_A	p_B	q_B
2017	4	5	3	3
2018	1	10	4	2

- What was nominal GDP in 2017?
- What was nominal GDP in 2018?
- What was real GDP in 2018 at 2017 prices (computed using fixed 2017 prices)? Using this measure, how much did GDP grow between 2017 and 2018?
- What was real GDP in 2017 at 2018 prices (computed using fixed 2018 prices)? Using this measure, how much did GDP grow between 2017 and 2018? What explains the difference between the two measures?
- How much did GDP grow between 2017 and 2018 using the chain-weighted method?

In Thailand, prices (in Thai baht) and quantities in 2017 were:

	p_A	q_A	p_B	q_B
2017	30	1	10	2

- What was nominal GDP in Thailand in 2017, expressed in baht.
- Suppose the exchange rate in 2017 was 25 baht per dollar. What was GDP in Thailand in 2017, expressed in US dollars at market exchange rates?
- What was GDP in Thailand in 2017 at PPP? What accounts for the difference between the market exchange rate measure and the PPP measure?
- What was the PPP exchange rate between the baht and the dollar?

Solution:

$$(a) \text{ GDP}_{2017}^{\text{nom}} = p_A^{17} q_A^{17} + p_B^{17} q_B^{17} = 4 \cdot 5 + 3 \cdot 3 = 20 + 9 = \boxed{29} \checkmark$$

$$(b) \text{ GDP}_{2018}^{\text{nom}} = 1 \cdot 10 + 4 \cdot 2 = 10 + 8 = \boxed{18} \checkmark$$

Nominal GDP *fell* by 38%. Hold that thought — it is about to be contradicted by both real measures.

- Value 2018 quantities at 2017 prices:

$$\text{GDP}_{2018}^{\text{real}} \Big|_{p^{17}} = 4 \cdot 10 + 3 \cdot 2 = 40 + 6 = \boxed{46}$$

Growth relative to 2017 nominal (which is also 2017 at 2017 prices, = 29):

$$g_L = \frac{46}{29} - 1 = 0.5862 = \boxed{58.62\%}$$

✓

(d) Value 2017 quantities at 2018 prices:

$$\text{GDP}_{2017}^{\text{real}} \Big|_{p^{18}} = 1 \cdot 5 + 4 \cdot 3 = 5 + 12 = \boxed{17}$$

Growth, comparing to 2018 nominal (= 18, i.e. 2018 at 2018 prices):

$$g_P = \frac{18}{17} - 1 = 0.0588 = \boxed{5.88\%}$$

✓

Why the two differ so much (58.6% vs 5.9%). Good A collapsed in relative price (from $p_A/p_B = 4/3$ to $1/4$) and simultaneously *doubled* in quantity. That is the textbook substitution pattern.

- Using **2017 prices** gives good A a weight of 4 — its old, high price. The good that grew fastest is weighted by the price it had *before* it got cheap, so growth looks large. This is the **Laspeyres** (base-weighted) direction and it **overstates** growth.
- Using **2018 prices** weights A at 1 — its new, low price. The fast-growing good barely counts, so growth looks small. This is the **Paasche** direction and it **understates** growth.

Neither is “right”. The gap between them is a genuine index-number problem, not an error.

(e) The chain-weighted (Fisher) growth rate is the geometric mean of the two links:

$$1 + g_C = \sqrt{(1 + g_L)(1 + g_P)} = \sqrt{1.5862 \times 1.0588} = 1.2960$$

$$g_C = \boxed{29.60\%}$$

✓

It sits between the two, as it must, and it is symmetric: it gives no privileged status to either year’s prices. This is why the BEA (and the IBGE) publish chained series.

(f) $\text{GDP}_{2017}^{\text{Thai}} = 30 \cdot 1 + 10 \cdot 2 = 30 + 20 = \boxed{50 \text{ baht}}$ ✓

(g) At the market rate of 25 baht per dollar:

$$\frac{50 \text{ baht}}{25 \text{ baht}/\$} = \boxed{\$2}$$

✓

(h) At PPP, value *Thai quantities* at *US prices* (the common price vector):

$$\text{GDP}^{\text{Thai, PPP}} = p_A^{\text{US}} q_A^{\text{Thai}} + p_B^{\text{US}} q_B^{\text{Thai}} = 4 \cdot 1 + 3 \cdot 2 = 4 + 6 = \boxed{\$10}$$

✓

The PPP measure is five times the market-exchange-rate measure. The reason: at the market rate, Thai goods are being valued at *Thai* prices converted to dollars, and Thai prices are low. Good A costs 30 baht = \$1.20 in Thailand but \$4 in the US; good B costs 10 baht = \$0.40 versus \$3. Thailand's output is physically the same whatever prices we use to value it, and PPP prices it consistently with US output. Market exchange rates are driven by the supply and demand for *tradables* and for financial assets, so they systematically understate the real output of poor countries, where non-tradables (housing, haircuts, domestic services) are cheap. This is the Balassa–Samuelson logic.

- (i) The PPP exchange rate is the rate that equalises the cost of the same basket:

$$e^{PPP} = \frac{\text{GDP in baht}}{\text{GDP in PPP dollars}} = \frac{50}{10} = \boxed{5 \text{ baht per dollar}}$$

✓

Compare with the market rate of 25 baht/\$. The baht is “undervalued” by a factor of 5 on this measure — equivalently, the price level in Thailand is one fifth of the US level.

```
pA17,qA17,pB17,qB17 = 4,5,3,3
pA18,qA18,pB18,qB18 = 1,10,4,2
n17 = pA17*qA17 + pB17*qB17      # 29
n18 = pA18*qA18 + pB18*qB18      # 18
r18 = pA17*qA18 + pB17*qB18      # 46  -> gL = 46/29-1 = 58.62%
r17 = pA18*qA17 + pB18*qB17      # 17  -> gP = 18/17-1 = 5.88%
gC  = ((r18/n17)*(n18/r17))*0.5 - 1 # 29.60%
th  = 30*1 + 10*2                  # 50 baht
ppp = pA17*1 + pB17*2              # $10 ; market: 50/25 = $2
```

Problem 1.3 Chained GDP

Question

The country of Fructus produces Apples, Bananas and Cherries. Its production statistics are given in the book's table for 2000–2020 (also downloadable as a spreadsheet from the book website).

- Compute a real GDP series at year-2000 prices using base year prices and using the chained method.
- Plot both series and comment on any differences.
- What was the average growth rate according to each method?

Solution:

- (a) Two constructions, both normalised to nominal GDP in 2000 (= 9,500):

Fixed base-year prices. $Y_t^{base} = \sum_i p_{i,2000} q_{i,t}$.

Chained (Fisher). Build year-on-year links and cumulate:

$$L_t = \frac{\sum_i p_{i,t-1} q_{i,t}}{\sum_i p_{i,t-1} q_{i,t-1}}, \quad P_t = \frac{\sum_i p_{i,t} q_{i,t}}{\sum_i p_{i,t} q_{i,t-1}}, \quad Y_t^{chain} = Y_{t-1}^{chain} \sqrt{L_t P_t}$$

Selected values (full series from the code below):

Year	Nominal	Base-2000	Chained
2000	9,500.0	9,500.0	9,500.0
2001	9,782.0	9,650.0	9,640.0
2005	10,021.0	10,400.0	10,341.8
2010	10,812.0	11,230.0	10,950.0
2015	12,048.0	12,390.0	11,922.8
2020	13,127.0	13,960.0	12,976.2

- (b) **The two series separate steadily, with the base-year series always above.**

By 2020 the fixed-base measure reads 13,960 against 12,976 for the chained one — a gap of about 7.6% that opens up gradually rather than jumping.

The mechanism is exactly the one flagged in Problem 1.2(d). Look at what happens to cherries: their price falls from 20 to 12 (a 40% decline) while their quantity rises from 200 to 361 (a +81% increase). Bananas do the opposite — price up from 30 to 61, quantity down from 50 to 34. Holding 2000 prices fixed keeps valuing cherries at their old, high price of 20 forever, so the fastest-growing good keeps an inflated weight and measured growth is too high. The chained index re-weights every year and tracks the moving relative prices.

- (c) Average annual growth over the 20 year span, $g = (Y_{2020}/Y_{2000})^{1/20} - 1$:

Method	2020 level	Average growth
Base-2000 prices	13,960.0	1.9432%
Chained	12,976.2	1.5713%
Difference		0.3718 p.p. per year

✓ Both computed in Python; code below.

Interpretation. A gap of 0.37 percentage points a year sounds small and is not: compounded over the two decades it is the entire 7.6% level discrepancy. This is the substitution bias in its purest form, and it is precisely why the US switched from fixed-base to chained real GDP in 1996 — the old method was overstating growth in exactly the sectors (computers, electronics) where prices were falling fastest and quantities rising fastest.

```

import numpy as np
q = np.array([qA, qB, qC], float).T      # (21, 3) quantities, 2000..2020
p = np.array([pA, pB, pC], float).T      # (21, 3) prices

base = (q * p[0]).sum(axis=1)             # fixed 2000 prices

chain = np.empty(len(q)); chain[0] = (q[0]*p[0]).sum()
for t in range(1, len(q)):
    L = (q[t]*p[t-1]).sum() / (q[t-1]*p[t-1]).sum() # Laspeyres link
    P = (q[t]*p[t]).sum() / (q[t-1]*p[t]).sum()     # Paasche link
    chain[t] = chain[t-1] * np.sqrt(L*P)            # Fisher link

n = len(q) - 1
print((base[-1]/base[0])** (1/n) - 1)           # 0.019432
print((chain[-1]/chain[0])** (1/n) - 1)          # 0.015713

```

Problem 1.5 Changes in Relative Prices

Question

We saw that using an earlier year as the base year to compute real GDP results in a high rate of growth if the sectors that are expanding most are those whose relative price is falling. Can you think of reasons why that should be the case (i.e., economic forces that make the same types of goods become relatively cheaper and be produced in higher quantities)? Can you think of reasons why the opposite should be the case (i.e., economic forces that make the same types of goods become relatively expensive and be produced in higher quantities)?

Solution:

The question is really about whether the observed price-quantity pair is generated by a **supply shift** or a **demand shift**. Sliding along a stable demand curve gives falling price with rising quantity; sliding along a stable supply curve gives rising price with rising quantity.

(a) **Cheaper and more of it — supply-driven (the common case).**

- **Sector-biased technological progress.** Productivity growth is wildly uneven across sectors. Semiconductors, computers, telecoms and increasingly solar panels have had enormous productivity gains, which push the supply curve out: unit cost and therefore relative price fall, and quantity rises. This is the single most important reason and the one that motivated the switch to chained indices.
- **Learning by doing and economies of scale.** Cumulative production lowers unit cost, which lowers price, which raises quantity, which lowers cost further — a self-reinforcing loop.

- **Elastic demand.** For the quantity response to be *large*, demand must be price-elastic. Goods with close substitutes and discretionary timing (consumer electronics) qualify; necessities do not.
- **Trade liberalisation and offshoring** shift the effective supply curve out for importables.

(b) **More expensive *and* more of it — demand-driven.**

- **Income-elastic goods (luxuries).** As income grows, demand for health-care, education, higher-end housing and personal services shifts out faster than supply. Both price and quantity rise.
- **Baumol's cost disease.** In sectors where productivity growth is inherently slow — a string quartet needs the same four players it needed in 1800 — wages still have to track the economy-wide average or workers leave. Unit costs, and hence relative prices, rise. If demand is income-elastic (health, education, live performance), quantity rises *too*. This is the cleanest example of the combination the question asks about.
- **Binding supply constraints.** Urban land, waterfront property, rare earths: the supply curve is steep, so a demand shift moves price much more than quantity — but quantity does still rise.
- **Network effects / quality ladders.** A good becomes more valuable as more people use it, shifting demand out while the price rises.

The index-number consequence. Base-year weighting overstates growth in case (1) and *understates* it in case (2), because there the expanding sectors are the ones whose prices are rising, so old prices under-weight them. Since case (1) empirically dominates in modern economies — technology-intensive goods are both the fastest-growing and the fastest-cheapening — fixed-base real GDP has an upward bias overall. Problem 1.3 is a constructed illustration of exactly this.

Connection to the course: the same asymmetry reappears in rules/07_moeda_inflacao.md as the **substitution bias in the CPI/IPCA** — a Laspeyres consumer price index overstates inflation for the same reason a Laspeyres quantity index overstates growth.

The Boundaries of GDP

Question

Read the following article: <http://www.economist.com/news/finance-and-economics/21603073-italys-inclusion-illicit-activities-its-figures-excit>

What do you think? Should drug production and prostitution be included in the calculation of GDP?

Solution:

Note on sourcing

This exercise sends you to a specific 2014 *Economist* piece, which I have not retrieved. The background it refers to is the EU's ESA 2010 accounting standard, which from September 2014 required member states to include illegal activities — drugs, prostitution, smuggling — in GDP where they are voluntary market transactions. Italy's measured GDP rose by roughly one percent as a result. The argument below is the economics; check the article for its specific framing before quoting it.

Position: yes, they should be included — and the reasons are about consistency, not about morality.

- (a) **The production boundary is defined by transactions, not by approval.** GDP already counts tobacco, alcohol, gambling, weapons and payday lending. It counts legal prostitution in the Netherlands and Germany and legal cannabis in Colorado. If the criterion were social desirability, the accounts would need a moral weighting for every line item, and no two countries would agree on it. The criterion actually used — was there a voluntary exchange producing something someone paid for — is at least well defined.
- (b) **Comparability collapses without it.** This is the strongest argument. Cannabis is legal in some EU countries and not others; prostitution is legal in Germany and illegal in France. If illegal activity is excluded, then Germany's GDP includes its sex industry and France's does not, and the two numbers are not measuring the same thing. Worse, a country that *legalises* an activity would show a spurious jump in GDP with no change in real activity. Uruguay's cannabis legalisation and Portugal's decriminalisation would both create artefacts. Including illegal activity everywhere removes the discontinuity.
- (c) **Coherence of the accounts.** The inputs to illegal industries are often already counted — the chemicals, the rent, the imported precursors. Counting inputs but not output puts a hole in the value-added identity.
- (d) **Policy relevance.** Debt-to-GDP ratios, EU budget contributions and deficit limits are all denominated in GDP. Italy's revision changed its measured debt ratio without changing a euro of debt. That is an argument for measuring the denominator consistently, not for leaving parts of it out.

The serious objections, and what they actually establish:

- (i) **Measurement is guesswork.** There are no invoices. Estimates are built from seizure data, survey-based prevalence and assumed prices, and revisions are large. *But* — this is an argument about *precision*, not about *principle*. Owner-occupied housing rent is also imputed rather than observed, and it is a far larger share of GDP than drugs.
- (ii) **Coercion breaks the “voluntary” premise.** Trafficking and coerced prostitu-

tion are not voluntary exchange, and treating them as consensual market output is a genuine category error. The honest response is that the accounts cannot separate the coerced from the voluntary component, which is a real and unresolved defect.

- (iii) **The welfare reading is wrong.** Counting drug sales as output while ignoring addiction, overdose and crime overstates wellbeing. This is correct — but it is not special to illegal goods. GDP does not net out pollution, commuting or the harm from any legal product either.

Conclusion. Objection (iii) is the deepest, and it does not argue for excluding illegal activity from GDP. It argues that **GDP is not a welfare measure and should not be read as one** — which is exactly the thesis of Chapter 2 and of the Jones–Klenow λ . Include the activity in the production measure; use a separate welfare measure when the question is whether people are better off.

Chapter 2 Beyond GDP

The question. GDP per capita measures production. Welfare depends on consumption, how it is distributed, how long people live and how much they work. Chapter 2 builds measures that price those things.

The HDI is a simple geometric average of three normalised sub-indices (income, education, life expectancy). It is transparent but the weights are arbitrary — there is no theory saying a year of life expectancy is worth a given amount of income.

The Jones–Klenow λ fixes exactly that by grounding the weights in preferences. Behind a Rawlsian veil of ignorance, ask: by what factor λ would we have to scale *US* consumption to make a person indifferent between being dropped at random into the *US* and into country *j*? That λ is an **equivalent variation** and is the country's welfare relative to the *US*. The book's eq. (2.2.1) is

$$u(c, l, a) = E \left[\bar{u} + \frac{c^{1-\sigma}}{1-\sigma} - \theta(1-l)^2 \right] a$$

and λ solves $u(\lambda c_{US}, l_{US}, a_{US}) = u(c_j, l_j, a_j)$.

The role of σ . The curvature parameter does double duty. It is risk aversion behind the veil, and because a concave u penalises spread, it is simultaneously the *inequality aversion* of the welfare measure. Higher σ means a more unequal country is marked down harder. As $\sigma \rightarrow \infty$ the criterion converges on Rawls's difference principle: only the worst-off person counts.

The empirical punchline. Western Europe looks *better* on λ than on GDP per capita (more leisure, longer lives, less inequality); much of Africa looks *worse*, driven mainly by low life expectancy.

The HDI and the Jones–Klenow Welfare Measure

Problem 2.1 Comparing the HDI and the Jones & Klenow Welfare Measure

Question

Download the UN data that goes into building the HDI from <http://hdr.undp.org/en/data> and the Jones & Klenow Data from <http://web.stanford.edu/~chadj/papers.html#rawls>. For each country, construct: (a) Relative GDP: $\frac{\text{GDP per capita}}{\text{GDP per capita in the US}}$; (b) Welfare-to-GDP: $\frac{\lambda}{\text{Relative GDP}}$; (c) HDI-to-GDP: $\frac{\text{HDI}}{\text{Relative GDP}}$; and plot a scatterplot of Welfare-to-GDP against HDI-to-GDP. What do each of these ratios measure? Is the impression we get from the Jones & Klenow measure very different from the one we get from the HDI? What countries look better under each measure and why?

Solution:

Data exercise — no closed-form answer

This requires two external datasets. No numbers are asserted below — the method, the expected pattern (which follows from the structure of the two measures and from the book's own Figure 2.2.3) and runnable code are given instead.

(a) What each ratio measures.

- **Relative GDP** is the pure income yardstick, normalised so the US = 1.
- **Welfare-to-GDP** = $\lambda / (\text{relative GDP})$ asks: *once we account for consumption share, inequality, leisure and mortality, does this country do better or worse than its income alone would suggest?* A ratio above 1 means the country punches above its income; below 1 means income flatters it.
- **HDI-to-GDP** asks the same question using the UN's weights instead of preference-based ones.

Both are **residual** measures — they strip out income and show what the non-income components add. That is what makes them comparable on a scatterplot.

(b) Do the two give the same impression? Partly — and the disagreements are systematic.

Expect a **positive but loose correlation**. Both load heavily on life expectancy,

so countries with long lives relative to income (much of Latin America, southern Europe) score above 1 on both, and countries with an HIV/AIDS-driven mortality deficit (South Africa, Botswana) score below 1 on both. That shared component drives the positive slope.

The disagreements come from what each measure includes that the other does not:

Component	In λ ?	In HDI?
Life expectancy	yes, priced by preferences	yes, arbitrary weight
Consumption level	yes (consumption, not GDP)	indirectly, via income
Inequality	yes , via σ	no
Leisure / hours	yes , via θ	no
Education	no	yes, one third of the index

Therefore:

- **Better under λ than under HDI:** *Western Europe*. Short working hours and low inequality are rewarded by λ and completely invisible to the HDI. This is Jones and Klenow's headline finding — France and Italy look much closer to the US on welfare than on income.
 - **Better under HDI than under λ :** countries with **strong schooling relative to consumption and high inequality** — several post-socialist and some Latin American economies. The HDI credits their educational attainment; λ ignores education entirely and penalises their income dispersion.
 - **Worse under λ :** high-inequality middle-income countries generally. **Brazil** and **South Africa** are the clearest cases: the HDI's income component uses the *mean*, while λ 's concave u effectively uses something closer to a certainty-equivalent, which is far below the mean when dispersion is large.
- (c) **The methodological point the exercise is driving at.** The HDI's weights are chosen by committee; λ 's are derived from a stated utility function and can be argued with on their own terms — you can disagree about σ , θ or $\bar{u}$ and recompute. That transparency is the real advantage, more than any particular ranking. Conversely λ omits education, which most people would say matters, so neither dominates.

```
import pandas as pd, matplotlib.pyplot as plt

hdi = pd.read_csv("hdi.csv")          # country, hdi
jk  = pd.read_csv("jones_klenow.csv") # country, lambda, gdp_pc

df = hdi.merge(jk, on="country").dropna()
us = df.loc[df.country == "United States", "gdp_pc"].iloc[0]

df["rel_gdp"]      = df.gdp_pc / us
df["welfare_ratio"] = df["lambda"] / df.rel_gdp
df["hdi_ratio"]    = df.hdi      / df.rel_gdp

ax = df.plot.scatter("hdi_ratio", "welfare_ratio", alpha=.6)
ax.axhline(1, ls=":", c="gray"); ax.axvline(1, ls=":", c="gray")
print(df[["country", "rel_gdp", "welfare_ratio", "hdi_ratio"]].
      .sort_values("welfare_ratio", ascending=False).head(15))
print(df[["welfare_ratio", "hdi_ratio"]].corr())
```

Problem 2.2 Other Things that Matter

Question

The welfare measure λ takes into account data on consumption levels, inequality, leisure and mortality.

- Name one other variable that might be an important determinant of welfare that is not included in standard GDP calculations.
- What data would you need in order to figure out how much weight to give to this variable? Describe how you would use that data to come up with the correct way to include the variable in question in the welfare calculation. What choices that people make might reveal how much they care about this variable? Don't worry too much about the feasibility of the data-collection procedure, but think carefully about why the observed choices would be informative.

Solution:

The Lista assigns this exercise directly. The grader will be looking for the **revealed-preference logic** in part (b), not for a list of nice-sounding variables in part (a). Spend your words on (b).

- Candidate: personal security — the risk of being a victim of violent crime.**

It is a good choice for Brazil specifically: homicide rates vary enormously across states and municipalities while incomes do not vary nearly as much, which is what

makes the weight estimable. Other defensible answers: air and water quality, commuting time, political freedom and civil liberties, job security, or the quality of social relationships. Any of these is fine *provided* you can describe in (b) how a choice would reveal its value.

Note that GDP treats crime perversely: spending on locks, private security, alarms and insurance *adds* to GDP, so a more dangerous country can look richer.

- (b) **How to price it: find a choice where people trade the variable against money.** Stated preference (“how much would you pay to be safe?”) is unreliable because nothing is at stake. Revealed preference uses actual decisions with real consequences.

Method 1 — hedonic pricing of housing (the workhorse).

Two otherwise identical apartments, one in a neighbourhood with a higher homicide rate, will not command the same rent. The rent gap capitalises the value of the safety difference. Estimate

$$\ln(\text{rent}_i) = \beta_0 + \beta_1 \text{crime}_i + \gamma' X_i + \varepsilon_i$$

where X_i controls for size, age, amenities, school quality and distance to work. Then $-\beta_1$ is the marginal willingness to pay for a unit reduction in crime risk, as a share of housing cost.

Why this is informative: a household choosing where to live is making a real, costly, repeated trade-off between money and safety. It has every incentive to get it right.

Identification problem, and the fix: crime is correlated with everything else that makes a neighbourhood bad, so β_1 picks up more than safety. You need variation in crime that is unrelated to other neighbourhood characteristics — for example a policing reform rolled out in some districts and not others, or the timing of a UPP-style intervention. Compare rents *within* the same neighbourhood before and after.

Method 2 — compensating wage differentials.

Dangerous occupations pay more, all else equal. Regressing wages on fatality risk across occupations, controlling for skill, gives the wage premium per unit of mortality risk and hence a **value of a statistical life**. This is the same logic Jones and Klenow use to price life expectancy, which is why it slots into their framework naturally.

Method 3 — migration.

People leaving a violent region for a lower-paying job elsewhere reveal a lower bound on what they will pay for safety.

- (c) **How to put it into the welfare calculation.** Extend the flow utility with a security argument s :

$$u(c, l, a, s) = \left[\bar{u} + \frac{c^{1-\sigma}}{1-\sigma} - \theta(1-l)^2 + \psi v(s) \right] a$$

and calibrate ψ so that the model's marginal rate of substitution matches the estimated one:

$$\left. \frac{\partial u / \partial s}{\partial u / \partial c} \right|_{\text{model}} = \underbrace{\text{WTP per unit of safety}}_{\text{from the hedonic regression}}$$

Then recompute λ with the extra term. Countries safer than the US get λ revised up; Brazil, with homicide rates many times the US level, would be revised **down**.

A caveat worth stating: if crime mainly *kills* people, its cost is already partly captured through life expectancy a , and adding $\psi v(s)$ on top would double count. The security term should be calibrated to capture fear, avoidance behaviour and non-fatal victimisation — the part that does *not* show up in mortality.

Problem 2.9 Risk Aversion and Welfare in Western Europe

Question

If instead $\sigma = 1$, Jones and Klenow had set a higher value of σ , what would this do to measured welfare in Western European countries? Explain.

Solution:

Answer: measured welfare in Western Europe would rise further relative to the US.

The reasoning has two steps.

- (a) **σ is simultaneously risk aversion and inequality aversion.** Behind the veil of ignorance, an individual faces a lottery over who they will be. A more concave u (higher σ) makes them dislike that lottery's spread more. Equivalently, the certainty-equivalent consumption

$$c^{CE} = \left((1 - \sigma) E \left[\frac{c^{1-\sigma}}{1-\sigma} \right] \right)^{\frac{1}{1-\sigma}}$$

falls further below the mean as σ rises, so unequal countries are penalised more heavily.

- (b) **Western Europe is markedly less unequal than the US.** Gini coefficients for disposable income run roughly 0.25–0.32 in France, Germany, Belgium and the Nordics against about 0.39 in the US. So raising σ imposes a *larger* penalty on the US than on Europe. Since λ is measured *relative to the US*, and the denominator is being marked down more than the numerator, European λ rises.

Reinforcing channel. Higher σ also raises the weight on the $\bar{u}$ term relative to the consumption term — that is, on *being alive* versus consuming. Western Europe's life expectancy exceeds the US, so this pushes the same way.

Second reinforcing channel. Europe's consumption *level* is below the US. With higher σ the utility function flattens at high c , so differences in consumption levels matter less at the margin and Europe's income shortfall is discounted — again favouring Europe. All three channels point the same way, which is why the conclusion is robust.

Limiting case as a check. As $\sigma \rightarrow \infty$ the measure converges on the Rawlsian difference principle: welfare is determined by the worst-off person alone (Problem 2.10 makes this precise). On that criterion Western European welfare states, with their higher consumption floor, would rank *well above* the US. The comparative static is just the continuous version of that limit. ✓

Valuing Life Expectancy and the Composition of Consumption

Problem 2.3 Healthcare

Question

When we calculate consumption, one of the (many) categories of consumption is healthcare services.

- (a) Look up how much healthcare is consumed in the United States per year for a recent year. State the total dollar amount and also what fraction of GDP and what fraction of consumption is accounted for by healthcare. A good place to look for this data is the National Income and Product Accounts (NIPA) at https://apps.bea.gov/iTable/index_nipa.cfm.
- (b) Suppose that we are calculating welfare in the style of Jones and Klenow, taking into account the impact of life expectancy on utility. Should we therefore subtract consumption of healthcare services from our measure of consumption? What do you think?

Solution:

Look this up yourself — NIPA Table 2.3.5

Navigate to NIPA **Table 2.3.5**, “Personal Consumption Expenditures by Major Type of Product”, and read off the *Health care* line under Services. Divide by GDP (Table 1.1.5, line 1) and by total PCE.

As an order of magnitude to sanity-check what you find: US health spending has been running around 17–18% of GDP on the broadest (national health expenditure) definition, and health care is roughly a fifth of personal consumption expenditure. The PCE health-care line is *narrower* than total national health expenditure — it excludes investment in medical structures and equipment and some public health activity — so expect a smaller number from Table 2.3.5 than the headline “18% of GDP” figure you will see quoted. Report whichever you use and say which definition it is.

- (a)
- (b) **No — do not subtract it. But the question is sharper than it looks, and the right answer is “it depends what health care buys”.**

The case for subtracting (the “intermediate good” view). Health care is arguably not something people want for its own sake. Nobody enjoys chemotherapy. It is an *input* into producing health and longevity. If the Jones–Klenow measure already rewards a country for its life expectancy a , then also counting the consumption that *bought* that life expectancy counts the same benefit twice — once as the output (longer life) and once as the input (the spending). On this view health care is like the intermediate goods netted out in Chapter 1.

Why that argument fails, or at least overreaches:

- **It proves far too much.** By the same logic you would subtract food (an input into not starving), housing (an input into shelter) and exercise. Essentially all consumption is an input into something. The consumption measure would collapse.
- **Health care delivers more than longevity.** A hip replacement, cataract surgery, insulin, dental work and pain management mostly improve the *quality* of life, not its length. That benefit appears *nowhere* in $u(c, l, a)$ — not in a , and not in c if you strip health care out. Subtracting would make the measure worse, not better.
- **The double counting is not symmetric across countries.** The US spends far more per unit of life expectancy than other rich countries. If anything this is an argument that US consumption is *overstated* as a welfare measure — i.e. that a dollar of US health spending buys less wellbeing than a dollar elsewhere — which is a statement about *prices and efficiency*, not about whether the category belongs in the accounts.

The defensible position. Keep health care in consumption, and recognise that the genuine problem is a **price-measurement** one, not a boundary one. If US health care is expensive rather than abundant, then deflating US health consumption by an appropriate price index — one that accounts for quality and outcomes — would reduce measured real US consumption. That is where the correction belongs. Netting the whole category out is a blunt instrument that throws away the quality-of-life benefits along with the double count.

Problem 2.4 Construction and the Armed Forces

Question

Suppose we compared welfare in two otherwise identical countries. Country A dedicates 50% of GDP to maintaining a large army. Country B dedicates 50% of GDP to building houses. In which of the two countries would welfare be higher, according to the measure used by Jones and Klenow. What do you think of this?

Solution:

- (a) **What the measure says: welfare is the same in both, and low in both.**

This is the key mechanical point. The Jones–Klenow λ is built on **consumption**, not GDP. Both countries devote 50% of output to something that is not current consumption:

- Country A: $G = 0.5Y$ on defence, so $C = 0.5Y$.
- Country B: $I = 0.5Y$ on housing, so $C = 0.5Y$.

Since the two countries are “otherwise identical” — same leisure, same life expectancy, same inequality — and consumption is identical, $\lambda_A = \lambda_B$. Both are measured as *worse off* than an otherwise identical country that consumed its whole output.

- (b) **What I think of this: the measure is right about A versus B, and wrong about the level — but for different reasons in each case, and the housing case is the more serious defect.**

On defence (country A), the measure is defensible. An army is not consumed and yields no direct utility. It may be *necessary* — protecting the country from conquest — but that is precisely the point: it is a cost of maintaining the ability to consume, not a benefit. A country that must spend half its output on defence genuinely is worse off than an identical country that faces no threat. Treating it as forgone consumption captures something real. This is the same logic as the “regrettables” discussion in the welfare literature: commuting, prisons and pollution abatement all absorb resources without generating utility.

On housing (country B), the measure is clearly deficient. Houses are durable capital that yields a **flow of housing services for decades**. In a single-period snapshot country B looks identical to country A, but this is a timing artefact:

- In the **next** period, B’s residents enjoy a larger stock of houses, and national accounts *do* count that as consumption via **imputed rent** on owner-occupied housing. So over a full horizon B is unambiguously better off.
- The defect is therefore **the static, one-period framing**, not the use of consumption. Investment today is consumption tomorrow. Comparing two countries by current consumption alone penalises the one that is building its future — the same objection one would make to ranking a country as “worse off” for having a high saving rate.

- (c) **The general lesson.** Any welfare measure built on a single period’s consumption mis-ranks economies that differ in their *investment* rates. The right fix is to evaluate the **present discounted value** of the consumption stream, or equivalently to compare steady states rather than transition points — which is exactly what the Solow model of Chapter 4 forces you to do, and what the Golden Rule (§4.3) is designed to answer.

Problem 2.5 Measuring the Value of Life Expectancy

Question

Would each of the following observations be useful in determining how much to weigh life expectancy in measured welfare? Explain.

- (a) Observing how life expectancy varies across people with different income levels within a country.
- (b) Observing how life expectancy varies across countries.
- (c) Observing how much people are willing to pay for funerals.
- (d) Observing the difference in house prices in neighborhoods with different murder rates.

Solution:

The criterion throughout: to price life expectancy we need a situation in which someone **chooses**, at a real cost, between money and mortality risk. Correlations that arise from something *other* than such a choice do not identify the weight.

- (a) **Partly useful, but badly confounded — causality runs both ways.**

Rich people do live longer. But the correlation cannot be read as “people buy longevity with income” because:

- **Reverse causality:** healthier people earn more. Illness reduces labour supply and productivity, so part of the gradient runs from health to income.
- **Common causes:** education, discount rates and family background drive both. A patient, educated person both saves more and smokes less.

There is a further, subtler problem: even a *causal* income–longevity gradient measures how much longevity people *get*, not how much they *value* it, unless we know they were deliberately trading one for the other.

Verdict: suggestive, not identifying. Would need an income shock unrelated to health — a lottery win, an inheritance, a pension reform — to be usable.

- (b) **Least useful of the four.**

Cross-country variation in life expectancy is driven by public health systems, sanitation, epidemiology, conflict and climate — essentially none of which is an individual’s willingness-to-pay decision. Nobody *chose* to be born in a country with a high under-five mortality rate. Confounding is at its worst here because rich countries differ from poor ones in every dimension at once.

Verdict: useful for *describing* the differences that the welfare measure needs to weigh, but useless for determining the *weight* itself. It is the data you plug in, not the data you calibrate with.

- (c) **Not useful, and it is a category error.**

Funeral spending is consumption by **the survivors**, not by the deceased. It measures willingness to pay for a ritual, for social status and for grief management. It says nothing about how much a living person values an additional year of their

own life — the quantity a in the utility function. Funeral spending would still be positive in a world where nobody valued longevity at all.

Verdict: no. This is the distractor in the list, and identifying it as such is the point of including it.

(d) **The most useful of the four — this is the right kind of data.**

House prices in high- versus low-murder-rate neighbourhoods reflect a **real, voluntary, costly choice** in which households trade money directly against mortality risk. That is exactly the trade-off we need to price. The method is hedonic regression:

$$\ln(\text{price}_i) = \beta_0 + \beta_1 \text{murder rate}_i + \gamma' X_i + \varepsilon_i$$

Converting the price differential into a per-unit-of-risk figure yields a **value of a statistical life**, which is precisely what calibrates $\bar{u}$ in the Jones–Klenow utility function.

The identification caveat, which you should state: murder rates correlate with school quality, amenities, employment access and neighbourhood income, so a naive β_1 is biased. Credible estimates exploit sharp variation — policing interventions, boundary discontinuities, or before/after comparisons within the same area.

Related and equally valid: compensating wage differentials across occupations with different fatality rates. Same logic, different market.

(d) > (a) > (b) > (c). The ordering tracks one thing: how close the observation is to an actual choice in which a person gave up money for a lower chance of dying.

Problem 2.6 The Value of Life Expectancy

Question

In the US, average consumption per capita is about \$35,000. Life expectancy is 79 years, so in the Jones and Klenow experiment, Rawls would have a 79% chance of being alive. In Bolivia, average consumption per capita is \$3,700 and life expectancy is 68 years. Assume that there is no inequality in either country so that everyone who is alive gets the average level of consumption. Assume the following utility function:

$$u(c, a) = \left[\bar{u} + \frac{c^{1-\sigma}}{1-\sigma} \right] a$$

with $\sigma = 0.5$ and $\bar{u} = 48.1$.

- (a) Suppose you made a US resident the following offer: you can buy a health plan that costs x dollars per year and will extend your lifetime by one year. What is the price x that would make them indifferent between getting the health plan or not?
- (b) How much would a Bolivian resident be willing to pay for such a plan?
- (c) Write down the special case of equation (2.2.1) that applies to this example and solve for λ for Bolivia. Don't replace any numbers yet, leave it in terms of a_{US} , a_{Bol} , c_{US} , c_{Bol} , σ and $\bar{u}$.
- (d) Replace the values of a_{US} , a_{Bol} , c_{US} , c_{Bol} , σ and $\bar{u}$ to find a value for λ . How does it compare to $\frac{c_{Bol}}{c_{US}}$? Why?

Solution:

Setup. With $\sigma = 1/2$ the consumption term simplifies usefully:

$$\frac{c^{1-\sigma}}{1-\sigma} = \frac{c^{1/2}}{1/2} = 2\sqrt{c}$$

so the **flow** utility is $\phi(c) \equiv \bar{u} + 2\sqrt{c}$ and total utility is $u(c, a) = \phi(c) a$.

	c	a	$\phi(c) = 48.1 + 2\sqrt{c}$
US	35,000	79	422.2657
Bolivia	3,700	68	169.7553

- (a) **The US resident pays x so that living 80 years on $c - x$ equals living 79 years on c :**

$$[\bar{u} + 2\sqrt{c_{US} - x}] \cdot 80 = [\bar{u} + 2\sqrt{c_{US}}] \cdot 79$$

Solve for x in closed form:

$$x = c_{US} - \left(\frac{1}{2} \left[\frac{79}{80} \phi(c_{US}) - \bar{u} \right] \right)^2$$

Numerically:

$$\frac{79}{80} \cdot 422.2657 = 416.9874, \quad \frac{1}{2}(416.9874 - 48.1) = 184.4437, \quad 184.4437^2 = 34,019.48$$

$$x_{US} = 35,000 - 34,019.48 = \$980.52 \quad \checkmark$$

That is about 2.80% of annual consumption.

- (b) **Same calculation for Bolivia**, with $68 \rightarrow 69$ years:

$$\left[\bar{u} + 2\sqrt{3700 - x} \right] \cdot 69 = 169.7553 \cdot 68$$

$$\frac{68}{69} \cdot 169.7553 = 167.2951, \quad \frac{1}{2}(167.2951 - 48.1) = 59.5976, \quad 59.5976^2 = 3,551.86$$

$$x_{Bol} = 3,700 - 3,551.86 = \$148.14 \quad \checkmark$$

Comparison and the economics. In *absolute* terms the American pays 6.6 times more — unsurprising, since they are 9.5 times richer. But as a *share of consumption*:

$$\frac{x_{US}}{c_{US}} = 2.80\%, \quad \frac{x_{Bol}}{c_{Bol}} = 4.00\%$$

The Bolivian gives up a *larger fraction* of consumption. Why? Because they start with a shorter life. Going from 68 to 69 years is a proportionally bigger gain ($1/68 = 1.47\%$) than 79 to 80 ($1/79 = 1.27\%$), and because u is concave in c but *linear* in a , the proportional gain in years is what matters. The absolute gap, meanwhile, reflects that a year of life is worth more when each year is spent consuming more.

- (c) **The special case of (2.2.1).** Drop leisure (not modelled here) and the expectation over inequality (assumed away). Equation (2.2.1) becomes

$$u(c, a) = \left[\bar{u} + \frac{c^{1-\sigma}}{1-\sigma} \right] a$$

and λ is defined by making Rawls indifferent between Bolivia and a rescaled US:

$$a_{US} \left[\bar{u} + \frac{(\lambda c_{US})^{1-\sigma}}{1-\sigma} \right] = a_{Bol} \left[\bar{u} + \frac{c_{Bol}^{1-\sigma}}{1-\sigma} \right]$$

Solving for λ :

$$\frac{(\lambda c_{US})^{1-\sigma}}{1-\sigma} = \frac{a_{Bol}}{a_{US}} \left[\bar{u} + \frac{c_{Bol}^{1-\sigma}}{1-\sigma} \right] - \bar{u}$$

$$\lambda = \frac{1}{c_{US}} \left\{ (1-\sigma) \left(\frac{a_{Bol}}{a_{US}} \left[\bar{u} + \frac{c_{Bol}^{1-\sigma}}{1-\sigma} \right] - \bar{u} \right) \right\}^{\frac{1}{1-\sigma}}$$

- (d) **Plugging in** $\sigma = 0.5$, $\bar{u} = 48.1$, $a_{US} = 79$, $a_{Bol} = 68$, $c_{US} = 35,000$, $c_{Bol} = 3,700$:

$$\frac{a_{Bol}}{a_{US}} = \frac{68}{79} = 0.86076, \quad \phi(c_{Bol}) = 169.7553$$

$$0.86076 \times 169.7553 - 48.1 = 146.1243 - 48.1 = 98.0243$$

$$\lambda_{c_{US}} = \left(\frac{98.0243}{2} \right)^2 = 49.0122^2 = 2,402.19$$

$$\lambda = \frac{2,402.19}{35,000} = 0.0686 \quad \checkmark$$

Comparison with relative consumption:

$$\frac{c_{Bol}}{c_{US}} = \frac{3,700}{35,000} = 0.1057 \quad \text{versus} \quad \lambda = 0.0686, \quad \frac{\lambda}{c_{Bol}/c_{US}} = 0.649$$

Why λ is well below relative consumption. Bolivian welfare is only 6.9% of US welfare even though Bolivian consumption is 10.6% of US consumption — roughly a third lower. The entire wedge comes from **life expectancy**: Bolivians live 68 years against 79, a factor of 0.861, and in this utility function years multiply the flow, so the mortality shortfall scales welfare down directly. Consumption alone *understates* how much worse off Bolivia is.

This is the central empirical message of Jones and Klenow and the reason λ is worth computing at all: for poor countries, and especially for those with an AIDS-driven or conflict-driven mortality deficit, GDP per capita is not merely an imperfect welfare proxy — it is systematically *too optimistic*.

```
from scipy.optimize import brentq
sig, ubar = 0.5, 48.1
flow = lambda c: ubar + c**(1-sig)/(1-sig)      # = ubar + 2*sqrt(c)
u      = lambda c, a: flow(c)*a
cUS, aUS, cBol, aBol = 35000, 79, 3700, 68

x_us   = brentq(lambda x: u(cUS-x, aUS+1) - u(cUS, aUS), 0, cUS-1)      # 980.52
x_bol  = brentq(lambda x: u(cBol-x, aBol+1) - u(cBol, aBol), 0, cBol-1) # 148.14

lam = brentq(lambda l: aUS*flow(l*cUS) - aBol*flow(cBol), 1e-9, 1)      # 0.068626
# closed form agrees to 6 dp:
lam_cf = ((1-sig)/cUS**(1-sig) * ((aBol/aUS)*flow(cBol) - ubar))**(1/(1-sig))
```

Risk Aversion, Inequality and the Veil of Ignorance

Question

The students of Concave University are all identical in their ability and preferences, which are well described by the function $u(c) = E \left[\frac{c^{1-\sigma}}{1-\sigma} \right]$. Two employers recruit CU graduates. Stabilis offers each graduate a fixed salary of \$50,000. Lotteris instead offers them a base salary of \$20,000 plus a bonus scheme that depends on their performance, which is entirely driven by luck. The bonus is either \$10,000 or \$100,000, with equal probability. The job itself is the same in both firms. About half the CU graduates choose to work for Stabilis and half prefer Lotteris, and they all say they found it hard to choose because both offers were similarly attractive. What value of σ is consistent with their decisions?

Solution:

- (a) **Translate the story into an indifference condition.** “They found it hard to choose because both offers were similarly attractive” means the graduates are **indifferent** — that is the whole content of the setup, and it is what pins down σ . Lotteris pays base + bonus, so the two outcomes are

$$20,000+10,000 = 30,000 \quad \text{or} \quad 20,000+100,000 = 120,000, \quad \text{each with probability } \frac{1}{2}$$

Note the expected pay at Lotteris is $\frac{1}{2}(30,000) + \frac{1}{2}(120,000) = \$75,000$, far above Stabilis’s \$50,000. Lotteris must pay a **risk premium** of \$25,000 to attract half the class — and the size of that premium is exactly what reveals σ .

- (b) **Set up the equation.**

$$\frac{50,000^{1-\sigma}}{1-\sigma} = \frac{1}{2} \cdot \frac{30,000^{1-\sigma}}{1-\sigma} + \frac{1}{2} \cdot \frac{120,000^{1-\sigma}}{1-\sigma}$$

The $\frac{1}{1-\sigma}$ factor cancels *provided* $\sigma \neq 1$ — but be careful with the sign, since $1-\sigma < 0$ when $\sigma > 1$. Safer to keep it and solve numerically. The condition says that \$50,000 is the **certainty equivalent** of the Lotteris lottery.

- (c) **Solve.** There is no closed form; solve numerically. Bracketing:

σ	$u(50,000)$	$E[u_{\text{Lotteris}}]$	Preferred
0.50	447.214	519.615	Lotteris
1.00	10.8198	11.0021	Lotteris
1.7964	-2.2732×10^{-4}	-2.2732×10^{-4}	indifferent
2.00	-2.0000×10^{-5}	-2.0833×10^{-5}	Stabilis

$$\boxed{\sigma \approx 1.796} \quad \checkmark$$

- (d) **Interpretation.** A σ of about 1.8 is squarely inside the range macroeconomists normally use — calibrations typically sit between 1 and 4, and Jones and Klenow themselves use $\sigma = 1$. So the CU students’ behaviour is entirely ordinary.

Two things worth noting:

- σ is a curvature parameter, and its magnitude is only interpretable relative to the size of the gamble. This is a very large gamble — a $4\times$ spread between outcomes — and it takes only moderate curvature to generate a \$25,000 premium on it.

- **The monotonicity is the useful check.** At low σ everyone takes the lottery (its mean is 50% higher); at high σ everyone takes the safe job. There is exactly one crossing, so σ is identified. If the problem had said *everyone* chose Stabilis we would only get an inequality, $\sigma > 1.796$.
- (e) **Connection to Chapter 2's real purpose.** This same σ is the inequality aversion in the welfare measure. A society where consumption is 30,000 for half the population and 120,000 for the other half is, on these preferences, exactly as good as one where everyone has 50,000 — even though mean consumption in the first is 75,000. The \$25,000 gap *is* the welfare cost of that inequality. Problem 2.8 pushes this further.

```
from scipy.optimize import brentq
import numpy as np
safe, lo, hi = 50000, 20000+10000, 20000+100000      # 50000 ; 30000 ; 120000

def gap(s):
    if abs(s-1) < 1e-12:
        return np.log(safe) - 0.5*(np.log(lo) + np.log(hi))
    f = lambda c: c**(1-s)/(1-s)
    return f(safe) - 0.5*(f(lo) + f(hi))

print(brentq(gap, 0.05, 5))      # 1.796394
```

Problem 2.8 Inequality and Risk Aversion

Question

Compare the following two countries. Both have a population of 100. Within each country, we label individuals in order of increasing consumption. In country A, the consumption of individual j is $c_A(j) = 100 + 8j$, while in country B the consumption of individual j is $c_B(j) = 200 + 4j$.

- Plot the consumption patterns of each country, with an individual's label j (which ranges from 1 to 100) on the horizontal axis and their consumption on the vertical axis.
- Compute per capita consumption in each country. [Note: if you want, you can approximate sums with integrals]
- Suppose the utility function in both countries is $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$. For what values of σ is expected utility higher in country A? Interpret your results.

Solution:

- Both are straight lines in j ; A is steeper and starts lower.**

	intercept	slope	$c(1)$	$c(100)$
Country A	100	8	108	900
Country B	200	4	204	600

They cross where $100 + 8j = 200 + 4j \Rightarrow 4j = 100 \Rightarrow j = 25$, at $c = 300$. So:

- the **poorest quarter** ($j < 25$) is better off in **B**;
- the **richest three quarters** ($j > 25$) are better off in **A**.

This crossing is the whole exercise: neither country dominates, so the ranking must depend on how much we care about the bottom — i.e. on σ .

(b) **Per capita consumption.** Using $\sum_{j=1}^{100} j = \frac{100 \cdot 101}{2} = 5050$, so $\bar{j} = 50.5$:

$$\bar{c}_A = 100 + 8(50.5) = 100 + 404 = \boxed{504}, \quad \bar{c}_B = 200 + 4(50.5) = 200 + 202 = \boxed{402}$$

✓ (The integral approximation $\int_0^{100}$ would give $\bar{j} = 50$ and hence 500 and 400 — close, and fine for the qualitative point.)

Country A is 25.4% richer on average, but far more unequal: standard deviations are 230.9 for A against 115.5 for B, exactly double, since the spread scales with the slope. A ranking based on mean consumption alone would pick A without hesitation. The point of the exercise is that this is not obviously right.

(c) **Country A is preferred for $\sigma < \sigma^* \approx 2.072$, and B above it.**

Compute $E[u] = \frac{1}{100} \sum_{j=1}^{100} \frac{c(j)^{1-\sigma}}{1-\sigma}$ for each:

σ	$E[u_A]$	$E[u_B]$	Preferred
0	504	402	A
0.25	138.715	118.743	A
0.50	43.5329	39.6651	A
0.75	18.4992	17.7631	A
1 (log)	6.08797	5.95172	A
1.5	-0.0993398	-0.10323	A
2	-2.70274×10^{-3}	-2.72994×10^{-3}	A
2.0721	equal		indifferent
3	-5.31529×10^{-6}	-4.11151×10^{-6}	B

$E[u_A] > E[u_B] \iff \sigma < 2.072$
✓

Interpretation — the trade-off is level against dispersion.

- At $\sigma = 0$ utility is linear, expected utility is just mean consumption, and A wins by construction ($504 > 402$). **No inequality aversion at all.**
- As σ rises, the concavity of u increasingly penalises A's wider spread. A's advantage in the mean is eroded.
- At $\sigma^* \approx 2.072$ the two exactly offset: A's 25.4% higher average consumption is precisely cancelled by its doubled dispersion.

- Beyond that, **B wins** — society would rather be poorer on average than face A's spread. As $\sigma \rightarrow \infty$ only the worst-off individual matters, and since $c_B(1) = 204 > 108 = c_A(1)$, B wins decisively. That limit is the Rawlsian difference principle, which Problem 2.10 develops.

The methodological point. Whether A or B is the better society is *not* an empirical question that the data can settle. Both countries' consumption distributions are fully observed. The ranking depends entirely on σ — a value judgement about how much weight to give the poor, smuggled in under the guise of a preference parameter. That is both the strength of the Jones–Klenow approach (the judgement is explicit and you can vary it) and its limitation (it cannot tell you which σ is right).

```
import numpy as np
from scipy.optimize import brentq
j = np.arange(1, 101)
cA, cB = 100 + 8*j, 200 + 4*j
print(cA.mean(), cB.mean())           # 504.0 402.0

def EU(c, s):
    return np.mean(np.log(c)) if abs(s-1) < 1e-12 else np.mean(c**(1-s)/(1-s))

print(brentq(lambda s: EU(cA, s) - EU(cB, s), 0.05, 3.0)) # 2.072082
```

Problem 2.10 Risk Aversion and the Difference Principle

Question

Suppose an individual is trying to evaluate a society from behind the veil of ignorance. He knows that he can either be rich or poor, with equal probability. Expected utility is $E[u(c)] = \frac{u(c_{Rich}) + u(c_{Poor})}{2}$ where $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$. Let's now define an indifference curve: for any utility level $\bar{U}$, the indifference curve plots all the combinations of c_{Rich} and c_{Poor} such that expected utility is $\bar{U}$.

- Take the equation $\bar{U} = \frac{1}{2} \frac{c_{Rich}^{1-\sigma}}{1-\sigma} + \frac{1}{2} \frac{c_{Poor}^{1-\sigma}}{1-\sigma}$ and solve for c_{Rich} .
- Plot a map of indifference curves for $\sigma = 0.5$ ($\bar{U} = 2, \sqrt{10}, 4$); $\sigma = 2$ ($\bar{U} = -1, -0.4, -0.25$); $\sigma = 5$; $\sigma = 11$. In all cases put c_{Poor} on the horizontal axis and c_{Rich} on the vertical axis, both ranging over $[0, 10]$. Note that whenever you set c_{Poor} to the left of $[2\bar{U}(1-\sigma)]^{\frac{1}{1-\sigma}}$ you should set $c_{Rich} = \infty$.
- Notice that as σ becomes large, the indifference curves start to look like right angles. Explain how this relates to Rawls's difference principle.

Solution:

(a) **Solve for c_{Rich} .** Starting from

$$\bar{U} = \frac{1}{2} \cdot \frac{c_{Rich}^{1-\sigma}}{1-\sigma} + \frac{1}{2} \cdot \frac{c_{Poor}^{1-\sigma}}{1-\sigma}$$

multiply by $2(1-\sigma)$:

$$2\bar{U}(1-\sigma) = c_{Rich}^{1-\sigma} + c_{Poor}^{1-\sigma}$$

$$c_{Rich} = \left[2\bar{U}(1-\sigma) - c_{Poor}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Where the asymptote comes from. The bracket must stay positive. For $\sigma < 1$ we need $c_{Poor}^{1-\sigma} < 2\bar{U}(1-\sigma)$; for $\sigma > 1$ both $1-\sigma$ and the utility values are negative and the inequality flips. Either way there is a critical value

$$c_{Poor}^{\min} = \left[2\bar{U}(1-\sigma) \right]^{\frac{1}{1-\sigma}}$$

at which $c_{Rich} \rightarrow \infty$: this is precisely the caveat the problem states. **Below that level of c_{Poor} , no amount of consumption given to the rich can reach utility $\bar{U}$** — the poor person's misery cannot be compensated.

(b) **The plots.** The code below reproduces all four panels. What to look for:

- $\sigma = 0.5$: gently bowed curves, close to straight lines. The vertical asymptote sits at $c_{Poor}^{\min} = (\bar{U})^2$, so at $\bar{U} = 2$ it is at $c_{Poor} = 4$. Substitution between rich and poor is easy: taking a little from the poor is readily offset by giving to the rich.
- $\sigma = 2$: noticeably more curved; the asymptote is at $c_{Poor}^{\min} = -1/(2\bar{U})$, e.g. 0.5 for $\bar{U} = -1$.
- $\sigma = 5$ and $\sigma = 11$: the curves collapse towards **right angles**. Over most of the range the indifference curve is essentially vertical (raising c_{Rich} barely helps) and then essentially horizontal.

(c) **Why right angles mean Rawls.**

Right-angled (L-shaped) indifference curves are **Leontief** preferences, $U = \min\{c_{Poor}, c_{Rich}\}$ up to a monotone transformation. For such preferences only the *smaller* argument matters: at the kink, increasing the larger one leaves utility unchanged.

Formally, the marginal rate of substitution along the curve is

$$\text{MRS} = -\frac{dc_{Rich}}{dc_{Poor}} = \left(\frac{c_{Rich}}{c_{Poor}} \right)^{\sigma}$$

Take any point with $c_{Rich} > c_{Poor}$, so the ratio exceeds 1. Then

$$\sigma \rightarrow \infty \implies \left(\frac{c_{Rich}}{c_{Poor}} \right)^{\sigma} \rightarrow \infty$$

The curve becomes vertical everywhere off the 45° line. Interpretation: **the planner would give up an unbounded amount of the rich person's consumption for one extra unit for the poor person.**

That is exactly **Rawls's difference principle**: social arrangements should be judged by the position of the *worst-off* member, and an inequality is justified only if it improves the position of the worst off. In this framework Rawls is not a separate ethical theory bolted on — it is the **limiting case of expected-utility maximisation behind the veil of ignorance as risk aversion goes to infinity.**

Why this matters for Chapter 2. It locates Jones and Klenow's choice of σ on a spectrum of ethical positions rather than treating it as a technical calibration:

$\sigma = 0$	pure utilitarian — maximise <i>mean</i> consumption, inequality irrelevant
$\sigma = 1$	log utility — Jones & Klenow's baseline; maximise the geometric mean
$\sigma \rightarrow \infty$	Rawlsian — maximise the consumption of the worst-off

Choosing σ is choosing a social welfare criterion. Problem 2.8 showed this has real bite: it flipped the ranking of two countries at $\sigma^* \approx 2.07$.

```
import numpy as np, matplotlib.pyplot as plt

def c_rich(c_poor, Ubar, sigma):
    br = 2*Ubar*(1-sigma) - c_poor**(1-sigma)
    with np.errstate(invalid="ignore"):
        out = np.where(br > 0, np.abs(br)**(1/(1-sigma)), np.inf)
    return out

panels = [(0.5, [2, np.sqrt(10), 4]), (2, [-1, -0.4, -0.25]),
          (5, [-0.25, -4/54, -4.0**-5]), (11, [-0.1, -4.0**10/10**11, -4.0**-10/10])]

fig, axes = plt.subplots(2, 2, figsize=(10, 9))
cp = np.linspace(0.01, 10, 2000)
for ax, (sig, Us) in zip(axes.ravel(), panels):
    for U in Us:
        ax.plot(cp, c_rich(cp, U, sig), lw=1.4, label=f"Ubar={U:.3g}")
    ax.set_xlim(0, 10); ax.set_ylim(0, 10)
    ax.plot([0, 10], [0, 10], ls=":", c="gray", lw=.8) # 45-degree line
    ax.set_title(f"sigma = {sig}"); ax.set_xlabel("c_Poor")
    ax.set_ylabel("c_Rich"); ax.legend(fontsize=7)
plt.tight_layout()
```

Chapter 3 Basic Facts about Economic Growth

What this chapter is for. Before building a model of growth, establish the facts it has to explain. All three exercises are **empirical** — they ask you to go and look.

The very long run. Living standards were roughly flat for millennia and then took off around 1800. Compounding does the work: 1.5% a year sounds trivial and multiplies income by 4.4 over a century.

The Kaldor facts. Over long horizons in developed economies: (1) Y/L grows at a roughly constant rate; (2) K/L grows at a roughly constant rate; (3) K/Y is roughly constant; (4) the return on capital is roughly constant; (5) factor shares are roughly constant. Facts 2 and 3 together imply fact 4, but it is stated separately because the behaviour of r matters independently. These five facts are the target the Solow model of Chapter 4 is built to hit.

Across countries. Initially-rich countries grow at similar, middling rates. Initially-poor countries show *enormous* dispersion — Korea and Botswana race ahead while Congo and Madagascar fall further behind. So there is **no absolute convergence** in the world sample. Whether there is convergence *within* more homogeneous groups is exactly what Exercise 3.3 asks.

Reading growth data. Always plot income in **logs**: then a constant growth rate is a straight line and the *slope* is the growth rate. Comparing slopes across countries is comparing growth; comparing heights is comparing levels.

All three exercises in this chapter are data exercises

Exercises 3.1–3.3 require downloading the Maddison Project Database and the Penn World Table. No numerical answers are asserted below. For each one I give the exact procedure, the pattern you should expect to find and why, and runnable code. Where I give an approximate historical figure it is flagged as approximate — the precise answer depends on the data release and the PPP benchmark year, which is itself part of the lesson.

Long-Run Growth and Historical Comparisons

Problem 3.1 The Past is a Foreign Country

Question

Find the data compiled by Bolt et al. (2018) at <https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2018>. Starting from 1800, look up the US GDP per capita at intervals of one decade. For each of these points in time, find the country that currently has the closest GDP per capita to the past US level. When did the US become richer than present-day India, present-day China, and present-day Portugal?

Solution:

- (a) **Procedure.** Download `mpd2018.xlsx`, sheet `Full data`. The variable you want is `cgdppc` (real GDP per capita in 2011 US dollars, PPP-adjusted), which is the one constructed for *cross-country level* comparisons — as opposed to `rgdnpapc`, which is built for comparisons *over time within* a country. Getting this choice right is the whole methodological point of the exercise: you are comparing the past US to the present of other countries, which is a level comparison across both time and space.

Then, for each decade $t \in \{1800, 1810, \dots\}$, find the country j minimising $|y_t^{US} - y_{\text{latest}}^j|$.

- (b) **Approximate answers — verify these against the data yourself.**

The threshold years are found by locating where the US series crosses each country's current level:

Present-day country	Approx. current level	US passed it around
India	\$6,000–7,000	the 1870s–1880s
China	\$13,000–16,000	the 1900s–1910s
Portugal	\$27,000–30,000	the 1940s–1950s

Treat these as orders of magnitude, not answers

China's level in particular moves fast — its GDP per capita roughly doubled in the decade before the 2018 release — so the crossing year is sensitive to the vintage. Report the year *and* the release you used.

- (c) **What the exercise is teaching.** Three things, in increasing order of importance:
- **The magnitude of modern growth.** The US of the 1880s — railroads, telegraphs, mass steel — had roughly the material living standard of India today. Two centuries of compounding at rates that look small in any given year produced a gap of more than an order of magnitude.
 - **Time and space are interchangeable measures of development.** “How rich is India?” and “how far back is the US equivalent?” are the same question.

This is what makes the title apt.

- **The comparison is cruder than it looks.** An 1880 American and a 2018 Indian with the same measured real income do not consume remotely the same bundle. The Indian has antibiotics, vaccines, mobile telephony and electricity; the American had none of them. Chained price indices (Chapter 1) cannot handle goods that did not exist — the **new-goods bias** — so this kind of comparison almost certainly *understates* how much better off the present-day poor country is. That caveat is worth stating in your answer.

```
import pandas as pd

mpd = pd.read_excel("mpd2018.xlsx", sheet_name="Full data")
us = mpd[mpd.countrycode == "USA"].set_index("year")["cgdppc"]

latest = (mpd.dropna(subset=["cgdppc"]).sort_values("year")
          .groupby("country").tail(1).set_index("country")["cgdppc"])

for decade in range(1800, 2011, 10):
    if decade in us.index and pd.notna(us[decade]):
        match = (latest - us[decade]).abs().idxmin()
        print(f"{decade} US = {us[decade]:8,.0f}    closest today: {match} "
              f"({latest[match]:,.0f})")

for target in ["India", "China", "Portugal"]:
    lvl = latest[target]
    crossed = us[us > lvl]
    print(f"US passed present-day {target} (${lvl:,.0f}) in {crossed.index.min()}")
```

The Kaldor Facts

Problem 3.2 The Kaldor Facts in Other Countries

Question

Look up GDP accounts for some countries other than the US (a good source is <https://www.rug.nl/ggdc/productivity/pwt/>). Reproduce Figures 3.2.1–3.2.3 using data from those countries (try a rich country, a middle-income country and a poor country). Do the Kaldor facts seem to hold for those countries as well?

Solution:

- (a) What to build, and from which PWT variables.

Kaldor fact	Plot	PWT 10.x variables
1. Y/L grows steadily	$\ln(Y/L)$ vs year	rgdpna / emp
2. K/L grows steadily	$\ln(K/L)$ vs year	rnna / emp
3. K/Y constant	K/Y vs year	rnna / rgdpna
5. Factor shares constant	labour share vs year	labsh

Use **rgdpna** (national-accounts real GDP) for *within-country over time*, not **cgdpo**. Fact 4 (the return on capital) is not directly observed; it can be inferred from **labsh** and K/Y via $r + \delta = \alpha \frac{Y}{K}$.

(b) **Suggested countries and what to expect.**

- **Rich (e.g. France, Japan, Canada).** The facts should hold reasonably well — this is the sample Kaldor generalised from. Expect a caveat: Japan's K/Y rose substantially through its high-investment era and its growth rate is clearly *not* constant (rapid to 1990, then flat), so “constant growth” holds only piecewise. Labour shares in most rich countries have **drifted down** since around 1980 — a well-documented violation of fact 5 and an active research area.
- **Middle-income (e.g. Brazil, Mexico, Thailand).** Expect the facts to hold *much* worse. Brazil is a good case precisely because it fails so visibly: rapid growth to about 1980, then the lost decade, then stagnation. $\ln(Y/L)$ is not a straight line by any reading. K/Y moves substantially.
- **Poor (e.g. Madagascar, Congo, Malawi).** Expect the facts to fail badly. $\ln(Y/L)$ may be *flat or falling* over decades. Data quality is also weakest here — a point worth making explicitly rather than treating the series as gospel.

(c) **The answer to the question as asked: the Kaldor facts are properties of economies on a balanced growth path, not laws of nature.**

They hold approximately for rich countries over long horizons — which is exactly the domain Kaldor proposed them for — and they **fail systematically for countries that are not on such a path**. That is not a defect of the data; it is informative. A country in the middle of a growth miracle or a growth disaster is *in transition*, and the Solow model (Chapter 4) predicts that transitional dynamics look precisely like this: K/Y rising as a country accumulates towards its steady state, Y/L growth decelerating as diminishing returns bite.

So the honest conclusion has two parts: the facts describe steady states well, and most of the world is not in a steady state. Chapter 5 takes this up directly when it asks whether capital accumulation alone can account for cross-country income differences.

```

import pandas as pd, numpy as np, matplotlib.pyplot as plt

pwt = pd.read_excel("pwt1001.xlsx", sheet_name="Data")
countries = ["France", "Brazil", "Madagascar"]      # rich / middle / poor

fig, axes = plt.subplots(3, 3, figsize=(13, 10), sharex=True)
for row, ctry in enumerate(countries):
    d = pwt[pwt.country == ctry].set_index("year")
    y_l = d.rgdpna / d.emp                # fact 1
    k_l = d.rnna / d.emp                 # fact 2
    k_y = d.rnna / d.rgdpna             # fact 3

    axes[row,0].plot(np.log(y_l)); axes[row,0].set_ylabel(f"{ctry}\nlog(Y/L)")
    axes[row,1].plot(np.log(k_l)); axes[row,1].set_ylabel("log(K/L)")
    axes[row,2].plot(k_y); axes[row,2].set_ylabel("K/Y")
    print(ctry, "labour share: first", d.labsh.dropna().iloc[0].round(3),
          "last", d.labsh.dropna().iloc[-1].round(3))
plt.tight_layout()

```

Growth Across Countries and Convergence

Problem 3.3 Within-Region Convergence

Question

Look up cross-country GDP data (a good source is <https://www.rug.nl/ggdc/productivity/pwt/>). For each country, compute average GDP-per-capita growth 1960–2014. Produce a scatterplot of growth against initial GDP per capita like Figure 3.3.1 but separately for countries in each of three regions of the world: Europe, Latin America and Africa. Is it the case in any of these regions that initially-poor countries have grown faster?

Solution:

The Lista assigns this exercise directly. Note that it is a **data exercise** — you will need to download the Penn World Table and actually produce three scatterplots. Budget time for it; it is not a paper-and-pencil question. Deliver the three plots plus the fitted slopes.

(a) **Procedure.** For each country with data in both years, compute the average

annual growth rate

$$g_i = \left(\frac{y_{i,2014}}{y_{i,1960}} \right)^{1/54} - 1$$

and plot g_i against $\ln y_{i,1960}$ (logs on the horizontal axis — this is the standard convention and makes the convergence slope interpretable). Fit a line to each regional subsample. **A negative slope is evidence of convergence.**

(b) **What you should find, and why.**

Region	Expected slope	Reading
Europe	clearly negative	strong convergence
Latin America	mildly negative or flat	weak / no convergence
Africa	flat or positive	no convergence; possible divergence

- **Europe converges, and convincingly.** In 1960 Portugal, Spain, Greece and Ireland were far poorer than Germany, France and the UK; by 2014 the gap had narrowed dramatically. Ireland is the extreme case. Add the post-1990 Eastern European countries and the negative slope is stronger still.
- **Latin America shows little.** Countries started at broadly similar middle-income levels and mostly stayed there — the region as a whole failed to converge on the US. Within-region dispersion did not narrow much; Venezuela moved sharply in the wrong direction.
- **Africa shows none, and possibly the reverse.** Botswana and Mauritius grew rapidly from low bases while Congo, Madagascar, Niger and the Central African Republic stagnated or shrank. Initial income predicts almost nothing.

(c) **Interpretation — this is the distinction between absolute and conditional convergence.** The exercise is engineered to make the point that *the world sample shows no absolute convergence, but sub-samples can.*

The Solow model predicts that each country converges to **its own** steady state k^* , which depends on its s , n , δ and technology. That gives **conditional convergence**: countries approach *their own* target, faster the further away they are. It gives *absolute* convergence only if all countries share the same steady state.

- Within **Europe** — similar institutions, similar saving and demographic rates, free movement of capital, technology and labour, and for many the disciplining effect of EU accession — the assumption of a *common* steady state is roughly defensible. So conditional convergence shows up as absolute convergence.
- Within **Africa**, steady states differ enormously: institutions, conflict, health, schooling, geography and policy vary far more than initial income does. Countries are converging, but to widely different targets, so nothing shows up in a plot against initial income alone.

This is the phenomenon of **convergence clubs**. Homogeneous groups converge

internally; the world as a whole does not.

- (d) **The caveat that earns marks.** Restricting attention to a region conditions on *observable* regional membership, but it is still not the same as controlling for s and n directly. A proper conditional-convergence test regresses

$$g_i = \beta_0 + \beta_1 \ln y_{i,1960} + \beta_2 \ln s_i + \beta_3 \ln(n_i + g + \delta) + \varepsilon_i$$

and asks whether $\beta_1 < 0$. The regional split is a crude, transparent proxy for that. Also beware **selection**: requiring data in both 1960 and 2014 drops exactly the countries whose statistical systems collapsed — typically the worst performers — which biases African growth *upward* and understates divergence.

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt

pwt = pd.read_excel("pwt1001.xlsx", sheet_name="Data")
pwt["y"] = pwt.rgdpna / pwt.pop

w = (pwt[pwt.year.isin([1960, 2014])])
    .pivot(index="country", columns="year", values="y").dropna()
w["g"] = (w[2014] / w[1960]) ** (1/54) - 1
w["ly60"] = np.log(w[1960])

regions = {
    "Europe": ["France", "Germany", "Italy", "Spain", "Portugal", "Greece", "Ireland",
               "Netherlands", "Belgium", "Austria", "Denmark", "Sweden", "Norway",
               "Finland", "Switzerland", "United Kingdom"],
    "Latin America": ["Brazil", "Argentina", "Chile", "Colombia", "Peru", "Mexico",
                      "Uruguay", "Venezuela (Bolivarian Republic of)", "Ecuador", "Bolivia (Plurinational State",
                      "Paraguay", "Costa Rica", "Panama", "Guatemala", "Honduras"],
    "Africa": ["Botswana", "Kenya", "Nigeria", "Ghana", "Senegal", "Cameroon", "Niger",
               "Malawi", "Zambia", "Zimbabwe", "Mozambique", "Madagascar", "Mauritius",
               "South Africa", "Central African Republic"],
}

fig, axes = plt.subplots(1, 3, figsize=(15, 4.5), sharey=True)
for ax, (name, members) in zip(axes, regions.items()):
    d = w[w.index.isin(members)]
    ax.scatter(d.ly60, d.g)
    b, a = np.polyfit(d.ly60, d.g, 1) # slope, intercept
    xs = np.linspace(d.ly60.min(), d.ly60.max(), 10)
    ax.plot(xs, a + b*xs, "r-")
    ax.set_title(f"{name}: slope = {b:+.4f}")
    ax.set_xlabel("log GDP per capita, 1960")
axes[0].set_ylabel("average growth 1960-2014")
plt.tight_layout()
```

Chapter 4 The Solow Growth Model

The assumptions on F (numbered as in §4.1, and you will need them by number in Problem 4.4):

- **4.1 Constant Returns to Scale:** $F(\lambda K, \lambda L) = \lambda F(K, L)$.
- **4.2 Positive Marginal Product:** $F_K > 0, F_L > 0$.
- **4.3 Diminishing Marginal Product:** $F_{KK} < 0, F_{LL} < 0$.
- **4.4 Inada Conditions:** $\lim_{K \rightarrow 0} F_K = \infty, \lim_{K \rightarrow \infty} F_K = 0$.
- **4.5 Population:** $L_{t+1} = (1 + n)L_t$, with n exogenous.

CRS is what lets us work per worker: $y = f(k)$ with $k = K/L$.

Mechanics (§4.2). $k_{t+1} = \frac{(1 - \delta)k_t + sf(k_t)}{1 + n}$, with steady state $sf(k^*) = (n + \delta)k^*$.

It is globally stable, and Inada guarantees it exists and is unique. **The steady state depends on s, n, δ — never on the initial k_0 .** This single fact answers Problems 4.1 and 4.2.

Golden Rule (§4.3). $f'(k_{gold}) = n + \delta$; with Cobb–Douglas, $s_{gold} = \alpha$.

Factor markets (§4.4). The results Problem 4.3 runs on:

$$r^K = F_K(k, 1), \quad w = F_L(1, 1/k), \quad \boxed{r = r^K - \delta} \quad (\text{eq. 4.4.12})$$

With Cobb–Douglas $Y = K^\alpha L^{1-\alpha}$:

$$r^K = \alpha k^{\alpha-1}, \quad w = (1 - \alpha)k^\alpha$$

so **high k means high wages and a low return on capital**, and vice versa. Profits are zero (Proposition 4.2) — all output goes to workers or to capital owners.

Technological progress (§4.5). Only with $A_{t+1} = (1 + g)A_t$ does the model deliver long-run growth in y , and it does so *by assumption*. Without it, the Solow model predicts no long-run growth in GDP per capita at all.

Comparative Statics — Shocks to Capital and Labour

Problem 4.1 An Earthquake

Question

Suppose an economy behaves according to the Solow growth model. It starts out at $t = 0$ at a steady state, with no technological progress and no population growth. Suppose an earthquake destroys half the capital stock. As a consequence of this, what would happen in the short run (i.e. immediately) and in the long run (i.e. once the economy reaches steady state) to: (a) GDP, (b) GDP per capita, (c) wages, (d) interest rates.

Solution:

The Lista uses this exercise almost verbatim, then adds a part (b) in which the population growth rate is *also* halved because people emigrate. That extension changes the long-run answer — see the note at the end of this solution.

Setup. $n = 0$, $g = 0$. The initial steady state satisfies $sf(k^*) = \delta k^*$. The earthquake sets $K \rightarrow K/2$ with L unchanged, so

$$k_0 = \frac{k^*}{2}$$

The organising principle: k^* depends only on s , n , δ and the production function — not on the initial condition. The earthquake changes the initial condition and nothing else. Therefore **every long-run answer is “returns to exactly where it was”**, and all the action is in the transition.

Variable	Short run	Long run	Because
(a) GDP	↓	back to original	$Y = F(K, L)$, K halved
(b) GDP per capita	↓	back to original	$y = f(k)$, k halved
(c) Wages	↓	back to original	$w = (1 - \alpha)k^\alpha$
(d) Interest rate	↑	back to original	$r = \alpha k^{\alpha-1} - \delta$

(a) **GDP falls immediately, but by less than half.** With Cobb–Douglas,

$$\frac{Y_{new}}{Y_{old}} = \left(\frac{K/2}{K} \right)^\alpha = \left(\frac{1}{2} \right)^\alpha$$

With $\alpha = 1/3$ this is $0.5^{1/3} = 0.794$, so GDP falls about 21% — not 50%. The reason is **diminishing returns** (Assumption 4.3): the capital destroyed was the *least* productive capital at the margin, and labour is unaffected.

In the long run k returns to k^* and Y returns to its original level.

(b) **GDP per capita: same as (a).** Since L is unchanged, $y = Y/L$ moves exactly in proportion to Y . Falls by $1 - 0.5^\alpha$ on impact, returns fully in the long run.

- (c) **Wages fall.** $w = F_L = (1 - \alpha)k^\alpha$ is *increasing* in k . Workers have less capital to work with, so labour’s marginal product falls:

$$\frac{w_{new}}{w_{old}} = \left(\frac{1}{2}\right)^\alpha$$

the same proportional drop as output — because with Cobb–Douglas the labour share is constant at $1 - \alpha$, so w and y always move together. Returns to the original level in the long run.

- (d) **Interest rates rise.** $r = r^K - \delta = \alpha k^{\alpha-1} - \delta$ is *decreasing* in k (Assumption 4.3 again). Capital is now scarce, so its marginal product is high:

$$r_{new}^K = \alpha \left(\frac{k^*}{2}\right)^{\alpha-1} = 2^{1-\alpha} \alpha (k^*)^{\alpha-1} = 2^{1-\alpha} r_{old}^K$$

With $\alpha = 1/3$ the rental rate rises by a factor $2^{2/3} = 1.587$, i.e. about 59%. Returns to the original level in the long run.

The transition, which is the economically interesting part. At $k_0 = k^*/2$ we have $k < k^*$, so

$$sf(k_0) > \delta k_0$$

— investment exceeds depreciation, and k *rises* back towards k^* . During this period the economy grows **faster than its long-run rate** (which is zero here). Growth is fastest immediately after the shock and slows as k approaches k^* .

Why this matters empirically. This is the Solow model’s explanation of the post-war German and Japanese “growth miracles”. Wartime destruction pushed k far below k^* ; the rapid growth that followed was not a change in those countries’ fundamentals but **convergence back to an unchanged steady state**. It is also the model’s prediction that natural disasters have no permanent effect on income — a prediction that holds up reasonably well for rich countries with intact institutions, and much less well where the disaster damages institutions or human capital, neither of which is in the model.

The Lista adds that people emigrate, halving the population growth rate n . Now the shock hits a *parameter*, not just the initial condition, so the long-run answer changes:

$$k^* = \left(\frac{s}{n + \delta}\right)^{\frac{1}{1-\alpha}} \implies n \downarrow \Rightarrow k^* \uparrow$$

GDP per capita ends up *higher* than before the earthquake, along with wages; the interest rate ends up *lower*. Total GDP is ambiguous in level terms and grows more slowly (n is lower). Note the exercise says $n = 0$ initially — if you read “halved” as applying to a strictly positive n , say so explicitly; if $n = 0$ then halving it changes nothing and the answer reverts to the base case. **State which reading you adopt** — this ambiguity is the kind of thing the rubric in `rules/00_estilo_avaliacao.md` treats as a residual risk.

Problem 4.2 The Black Death

Question

In the middle of the 14th century, a plague killed about a third of the population of Europe. Assume that the economy of Europe was well described by the Solow model. How would the following variables change in response to the Black Death in the short run (i.e. immediately)? (a) GDP, (b) GDP per capita, (c) wages, (d) interest rates.

Solution:

Setup. $L \rightarrow \frac{2}{3}L$ with K unchanged. Therefore

$$k = \frac{K}{L} \rightarrow \frac{K}{(2/3)L} = \frac{3}{2}k$$

Capital per worker rises by 50%. This problem is the mirror image of 4.1: there capital was destroyed and k fell; here labour is destroyed and k rises. Every sign flips.

Variable	Short run	With $\alpha = 1/3$
(a) GDP	↓	$\times (2/3)^{1-\alpha} = 0.763$ (−24%)
(b) GDP per capita	↑	$\times (3/2)^\alpha = 1.145$ (+15%)
(c) Wages	↑	$\times (3/2)^\alpha = 1.145$ (+15%)
(d) Interest rate	↓	$r^K \times (3/2)^{\alpha-1} = 0.874$ (−13%)

(a) **GDP falls.** Fewer workers, same capital:

$$\frac{Y_{new}}{Y_{old}} = \left(\frac{2}{3}\right)^{1-\alpha}$$

With $\alpha = 1/3$: $(2/3)^{2/3} = 0.763$, so total output falls about 24%. Note it falls by *less* than the third of the population that died — diminishing returns again, this time to labour.

(b) **GDP per capita rises.**

$$\frac{y_{new}}{y_{old}} = \left(\frac{k_{new}}{k_{old}}\right)^\alpha = \left(\frac{3}{2}\right)^{1/3} = 1.145$$

about +15%. This is the striking result and the point of the exercise: **the survivors were, on average, materially better off.** Output fell by less than population did, so output per head rose.

(c) **Wages rise, by the same 15%.** $w = (1 - \alpha)k^\alpha$ and k rose. Labour became the scarce factor and its marginal product rose accordingly.

This is the model's most successfully tested prediction. Real wages in England roughly doubled over the century following 1348, and the historical record shows exactly the pattern the model implies: acute labour scarcity, landlords

competing for tenants and labourers, and legislative attempts to suppress the adjustment — the English Statute of Labourers of 1351 tried to fix wages at pre-plague levels and largely failed. That legislation is itself evidence that wages were rising sharply.

(d) **Interest rates fall.** $r^K = \alpha k^{\alpha-1}$ is decreasing in k :

$$\frac{r_{new}^K}{r_{old}^K} = \left(\frac{3}{2}\right)^{\alpha-1} = \left(\frac{3}{2}\right)^{-2/3} = 0.874$$

Capital became abundant relative to labour, so its return fell. Since $r = r^K - \delta$, the interest rate falls by the same absolute amount as the rental rate. Land rents fell for the same reason — another well-documented feature of the post-plague period.

The question asks only about the short run, but the long run is worth a sentence. With n unchanged, population recovers, k falls back to k^* , and **wages and income per head return to their original levels**. The gains to survivors were transitory. That is roughly what happened historically over the following two centuries — and it is precisely the **Malthusian** mechanism of Problem 4.5, where population growth responds endogenously to living standards. In the Solow model with exogenous n the return to k^* is mechanical; in the Malthusian model it is *driven* by the high living standards themselves. Problem 4.5 makes that explicit.

	4.1 Earthquake ($K \downarrow$)	4.2 Plague ($L \downarrow$)
$k = K/L$	falls	rises
GDP	falls	falls
GDP per capita	falls	rises
Wages	falls	rises
Interest rate	rises	falls

Both shocks reduce total output. They have **opposite** effects on everything per-capita and on both factor prices, because they move k in opposite directions. If you can state *why* — factor prices are determined by k alone, via $w = (1 - \alpha)k^\alpha$ and $r^K = \alpha k^{\alpha-1}$ — you have understood §4.4.

Factor Markets and Factor Prices

Question

Suppose both North and South Korea have the same technology level (!?), which doesn't grow, and also have constant population. Their respective populations and capital stocks are $L_{South} = 10$, $K_{South} = 2430$, $L_{North} = 5$, $K_{North} = 160$. The production function is $Y = K^\alpha L^{1-\alpha}$ with $\alpha = 0.4$. The depreciation rate is $\delta = 0.08$.

- Compute GDP per capita in each country.
- Compute wages and interest rates in each country.
- Suppose North and South Korea unify. What is the new $\frac{K}{L}$ in the unified country? What is GDP per capita in the unified country?
- Compute wages and interest rates in the unified country. Interpret.
- Will people and/or machines physically move between the North and the South? In what direction?

Solution:

Formulas. With $Y = K^{0.4}L^{0.6}$:

$$y = f(k) = k^{0.4}, \quad w = F_L = (1 - \alpha)k^\alpha = 0.6 k^{0.4},$$

$$r^K = F_K = \alpha k^{\alpha-1} = 0.4 k^{-0.6}, \quad r = r^K - \delta = r^K - 0.08$$

- (a) **GDP per capita.** The numbers are chosen to come out exactly:

$$k_S = \frac{2430}{10} = 243 = 3^5 \implies y_S = 243^{0.4} = (3^5)^{0.4} = 3^2 = \boxed{9}$$

$$k_N = \frac{160}{5} = 32 = 2^5 \implies y_N = 32^{0.4} = (2^5)^{0.4} = 2^2 = \boxed{4}$$

✓ Total output: $Y_S = 9 \times 10 = 90$ and $Y_N = 4 \times 5 = 20$.

The South is 2.25 times richer per head, purely because it has 7.6 times more capital per worker.

- (b) **Wages and interest rates.**

	k	$w = 0.6 k^{0.4}$	$r^K = 0.4 k^{-0.6}$	$r = r^K - 0.08$
South	243	$0.6 \times 9 = \mathbf{5.4}$	$0.4/27 = \mathbf{0.01481}$	$\mathbf{-0.0652}$
North	32	$0.6 \times 4 = \mathbf{2.4}$	$0.4/8 = \mathbf{0.05000}$	$\mathbf{-0.0300}$

✓ Exact: $243^{-0.6} = 3^{-3} = 1/27$ and $32^{-0.6} = 2^{-3} = 1/8$.

Southern workers earn $2.25 \times$ Northern workers; Northern capital earns $3.4 \times$ Southern capital. That is the whole economics of the problem: each factor is better paid where it is scarcer.

Both interest rates are negative — say so

$r_S = -6.52\%$ and $r_N = -3.00\%$. This is a property of the numbers the problem chose, not an error: $\delta = 0.08$ exceeds the marginal product of capital in *both* economies. Economically it means both are over-capitalised — they hold more capital than is worth maintaining, and are dynamically inefficient in the sense of §4.3. Neither would be in a steady state with a sensible saving rate. Flag this rather than quietly reporting a negative number.

(c) **Unification.** Capital and labour pool:

$$k_U = \frac{K_S + K_N}{L_S + L_N} = \frac{2430 + 160}{10 + 5} = \frac{2590}{15} = \boxed{172.67}$$

$$y_U = 172.67^{0.4} = \boxed{7.850}$$

✓

Note k_U is the **labour-weighted average** of k_S and k_N : $\frac{10(243)+5(32)}{15} = 172.67$.

✓

Total output rises. $Y_U = 7.850 \times 15 = 117.75$, against $90 + 20 = 110$ before — a gain of 7.75, about 7%, from nothing but reallocation.

Why is this not a violation of constant returns to scale? CRS says that scaling *both* inputs by λ scales output by λ . Here we are not scaling: we are **pooling two economies with different capital–labour ratios**. Since f is strictly concave (Assumption 4.3), Jensen’s inequality gives

$$f\left(\frac{L_S k_S + L_N k_N}{L_S + L_N}\right) > \frac{L_S f(k_S) + L_N f(k_N)}{L_S + L_N}$$

$$7.850 > \frac{90 + 20}{15} = 7.333 \quad \checkmark$$

Equalising k across a given total of capital and labour always raises total output, and the gain is larger the more unequal the initial ratios. This is the general case for factor mobility.

(d) **Factor prices in the unified country, and who wins.**

$$w_U = 0.6 \times 7.850 = \boxed{4.710}, \quad r_U^K = 0.4 \times 172.67^{-0.6} = \boxed{0.01819}, \quad r_U = \boxed{-0.0618}$$

✓

	Before	After	Effect
Southern workers	5.400	4.710	lose −12.8%
Northern workers	2.400	4.710	gain +96.3%
Southern capital (r^K)	0.01481	0.01819	gain +22.8%
Northern capital (r^K)	0.05000	0.01819	lose −63.6%

Interpretation. Unification is a Kaldor–Hicks improvement — total output rises by 7% — but it is emphatically **not** a Pareto improvement. There are clear winners and losers, and they are split *by factor, not by country*:

- **Winners:** Northern *workers* (their wages nearly double) and Southern *capital owners*.
- **Losers:** Southern *workers* and Northern *capital owners*.

This is the standard political economy of economic integration and of immigration: the scarce factor in each region loses when integration makes it less scarce. It explains why South Korean workers might rationally oppose a unification that raises aggregate Korean output — and why the aggregate gain, being real, could in principle be used to compensate them.

(e) **Direction of movement: capital flows South $\rightarrow$ North, labour flows North $\rightarrow$ South.**

- **Machines move North.** Capital chases the higher return: $r_N^K = 0.050 > r_S^K = 0.0148$. An investor earns $3.4\times$ more per unit of capital in the North.
- **People move South.** Labour chases the higher wage: $w_S = 5.4 > w_N = 2.4$.

Both flows do the same thing — they **equalise** k — and both stop exactly when $k_S = k_N = 172.67$, at which point w and r^K are equal everywhere and there is no further incentive to move. The model is **indifferent between the two channels**: only k matters, so whether adjustment happens through migration, through capital flows, or through any mix of the two, the final allocation and all factor prices are identical.

What the model leaves out. In reality the two channels are not interchangeable. Capital moves faster and more cheaply than people; migration has social and political costs the model does not price; and North Korean workers would not in fact be equally productive with Southern capital, because human capital and institutions differ — which is exactly the gap Chapter 5 fills with **TFP differences**. The German reunification experience, where massive capital transfers to the East did *not* equalise productivity, is the obvious cautionary case.

```
al, de = 0.4, 0.08
f = lambda k: k**al
w = lambda k: (1-al)*k**al
rK = lambda k: al*k**(al-1)

for name, K, L in [("South",2430,10), ("North",160,5), ("Unified",2590,15)]:
    k = K/L
    print(f"{name:8s} k={k:9.4f} y={f(k):7.4f} Y={f(k)*L:9.4f} "
          f"w={w(k):7.4f} rK={rK(k):8.5f} r={rK(k)-de:+.5f}")
# South    k= 243.0000 y= 9.0000 Y= 90.0000 w= 5.4000 rK= 0.01481 r=-0.06519
# North    k=  32.0000 y= 4.0000 Y= 20.0000 w= 2.4000 rK= 0.05000 r=-0.03000
# Unified  k= 172.6667 y= 7.8502 Y= 117.7537 w= 4.7101 rK= 0.01819 r=-0.06181
# gain from pooling: 117.7537 - 110 = +7.7537 (Jensen: f is strictly concave)
```

Alternative Production Technologies

Problem 4.4 An AK Model

Question

Suppose the production function takes the form $F(K, L) = AK$.

- (a) Which of the assumptions that we made about F does this satisfy and which does it not satisfy?
- (b) Suppose every other assumption we used in the Solow model holds, and there is no technological progress. Find an expression for $\frac{k_{t+1}}{k_t}$ (it may be useful to follow the steps that lead to expression (4.2.2)).
- (c) Will this economy grow in the long run? Explain.

Solution:

- (a) **Checking the assumptions of §4.1 one by one.**

Assumption	Status	Why
4.1 CRS	satisfied	$F(\lambda K, \lambda L) = A\lambda K = \lambda AK = \lambda F(K, L)$
4.2 Positive MP	fails	$F_K = A > 0 \checkmark$, but $F_L = 0$. Labour has <i>zero</i> marginal product — it is not a factor of production at all
4.3 Diminishing MP	fails	$F_{KK} = 0$, not < 0 . The marginal product of capital is constant at A forever. ($F_{LL} = 0$ too.)
4.4 Inada	fails, both	$\lim_{K \rightarrow 0} F_K = A \neq \infty$ and $\lim_{K \rightarrow \infty} F_K = A \neq 0$

Only CRS survives, and Assumption 4.5 (exogenous population growth) is untouched since it says nothing about F . The failure of **4.3** is the one that drives everything below — the absence of diminishing returns to capital is the entire content of the AK model.

- (b) **Deriving k_{t+1}/k_t .** Follow the standard steps. Per worker,

$$y_t = \frac{Y_t}{L_t} = \frac{AK_t}{L_t} = Ak_t$$

so $f(k) = Ak$ — a straight line through the origin rather than a concave curve. The capital accumulation equation is unchanged:

$$K_{t+1} = (1 - \delta)K_t + sY_t$$

Divide by $L_{t+1} = (1 + n)L_t$:

$$k_{t+1} = \frac{(1 - \delta)k_t + sf(k_t)}{1 + n} = \frac{(1 - \delta)k_t + sAk_t}{1 + n}$$

The k_t factors out:

$$\frac{k_{t+1}}{k_t} = \frac{1 - \delta + sA}{1 + n}$$

This is the key result: the growth factor is a constant, completely independent of k_t . Contrast with the standard Solow model, where $\frac{k_{t+1}}{k_t} = \frac{(1-\delta)+sf(k_t)/k_t}{1+n}$ and the term $f(k)/k$ is *decreasing* in k — that decline is what produces convergence to a steady state. Here $f(k)/k = A$ is constant, so the mechanism that generates a steady state is gone.

- (c) **Yes, this economy grows forever — provided sA is large enough — and this is *endogenous* growth.**

The economy grows if and only if $\frac{k_{t+1}}{k_t} > 1$:

$$\frac{1 - \delta + sA}{1 + n} > 1 \iff 1 - \delta + sA > 1 + n \iff \boxed{sA > n + \delta}$$

If this holds, k (and hence $y = Ak$) grows at the **constant rate**

$$g_k = \frac{1 - \delta + sA}{1 + n} - 1 = \frac{sA - n - \delta}{1 + n}$$

forever, with **no steady state and no convergence**. If $sA < n + \delta$ the economy shrinks to zero instead. There is no interior rest point either way.

Why the Inada conditions mattered. In standard Solow, $\lim_{K \rightarrow \infty} F_K = 0$ guarantees that accumulation eventually runs into diminishing returns: each new machine adds less, until investment only just covers depreciation and population growth. **Remove diminishing returns and accumulation never runs out of steam.**

The economically important consequence: the saving rate affects the long-run growth rate, not just the level.

$$\frac{\partial g_k}{\partial s} = \frac{A}{1 + n} > 0$$

This is the sharpest possible contrast with Solow, where s affects only y^* and has *no* effect on long-run growth (the “level effect versus rate effect” distinction from §4.2). A permanent increase in the saving rate here permanently raises growth. That makes policy — anything that raises s or A — permanently consequential, which is why the AK structure became the workhorse of the first generation of endogenous growth theory.

Is it plausible? The obvious objection is that $F_L = 0$ is absurd: labour clearly matters. The standard defence is that K should be read broadly, as *physical plus human capital plus knowledge*, and that the constant marginal product proxies for **externalities** — one firm’s capital raises other firms’ productivity, so the *social* return to capital does not diminish even though the private return does. The model is best read as a reduced form for that idea rather than as a literal technology.

```
A, s, delta, n = 0.5, 0.3, 0.08, 0.01
ratio = (1 - delta + s*A) / (1 + n)      # 1.05940594 -- constant

k = 1.0
for t in range(200):
    k = ((1-delta)*k + s*A*k) / (1+n)
print(k, 1.0*ratio**200)    # 102915.498144 102915.498144 -> exactly geometric

print(s*A, delta + n + n*delta)        # 0.15 > 0.0908 -> grows
print(((1-delta+0.4*A)/(1+n)) - 1)     # s: 0.3->0.4 raises growth 5.94% -> 10.89%
```

Problem 4.5 A Malthusian Model

Question

Look up Thomas Malthus and read a little about his work. We are going to formalize a simple version of his ideas in a little model. The production function is

$$Y_t = A_t L_t^\alpha \quad (4.5.5)$$

where A_t is technology, L is labor supply and $\alpha < 1$. There is no capital, no saving and no investment, so consumption is given by $C_t = Y_t$ (4.5.6). The population evolves according to

$$L_{t+1} = L_t \left(1 + \gamma \left(\frac{C_t}{L_t} - c \right) \right) \quad (4.5.7)$$

where γ and c are parameters.

- Malthus had in mind an agricultural economy where the total amount of land is fixed. What does that have to do with the way we wrote down the production function? What is the significance of assuming $\alpha < 1$?
- What does equation (4.5.7) mean? What is c ? What is γ ? Why did Malthus think that something like equation (4.5.7) applied?
- Assume $A_t = A$ is constant over time. Using (4.5.5), (4.5.6) and (4.5.7), find an expression for the percentage rate of population growth $\frac{L_{t+1}}{L_t}$ as a function of the level of population L_t . Plot this function.
- Assume $A_t = A$ is constant over time. Will GDP per capita grow in the long run? Will the population grow in the long run?
- Suppose there is a one-time increase in A_t , from A to $A' > A$. What will happen to GDP per capita and the size of the population in the short run and in the long run? Show your reasoning graphically and/or algebraically.
- Suppose now that instead of being constant, technology improves at a constant rate, so that $A_{t+1} = A_t(1 + g)$. Will GDP per capita grow in the long run? Will the population grow in the long run? Explain.
- When we have a constant rate of technological progress, how does the long-run level of consumption per capita depend on γ ? Why?

Solution:

- Land is the hidden fixed factor, and $\alpha < 1$ is what encodes it.**

Write the “true” technology with land T explicitly and constant returns in (labour, land): $Y = \tilde{A} L^\alpha T^{1-\alpha}$. If land is **fixed** at $\bar{T}$, absorb it into the constant:

$$Y = \underbrace{\tilde{A} \bar{T}^{1-\alpha}}_{\equiv A} L^\alpha$$

which is exactly (4.5.5). So the reduced-form production function *is* the constant-returns technology with the fixed factor substituted out.

Significance of $\alpha < 1$: it delivers **diminishing returns to labour**.

$$\frac{\partial Y}{\partial L} = \alpha AL^{\alpha-1} > 0, \quad \frac{\partial^2 Y}{\partial L^2} = \alpha(\alpha - 1)AL^{\alpha-2} < 0$$

and, critically, output per worker

$$\frac{Y_t}{L_t} = AL_t^{\alpha-1}$$

is **decreasing** in L because $\alpha - 1 < 0$. More people working the same fixed land means less output each. **This is the engine of the entire model** — without it ($\alpha = 1$) output per head would be independent of population and nothing that follows would work.

- (b) **Equation (4.5.7) makes population growth endogenous, responding to living standards.**

Rewrite it as

$$\frac{L_{t+1} - L_t}{L_t} = \gamma \left(\frac{C_t}{L_t} - c \right)$$

so the population *growth rate* is proportional to the excess of consumption per capita over c .

- **c is the subsistence level of consumption.** If $C/L > c$, the population grows; if $C/L < c$, it shrinks; if $C/L = c$, it is constant. It is the living standard at which the population exactly reproduces itself.
- **γ measures the responsiveness of population growth to living standards** — the semi-elasticity of demographic behaviour with respect to income. A high γ means population reacts strongly and quickly; a low γ means it is sluggish.

Why Malthus believed it. Writing in 1798, he had in mind two mechanisms:

- **Positive checks** operating below subsistence — famine, disease and war raise the death rate, especially infant and child mortality.
- **Preventive checks** operating above it — prosperity brings earlier marriage and higher fertility; hardship delays marriage and reduces it.

This was an accurate description of pre-industrial demography, where fertility genuinely did respond to harvests and real wages. Note the sharp contrast with Kurlat's **Assumption 4.5**, which makes n exogenous — the book flags in §4.1 that this is “actually a very big deal”, and this exercise is where you see why.

- (c) **The population growth function.** Substitute (4.5.5) and (4.5.6):

$$\frac{C_t}{L_t} = \frac{Y_t}{L_t} = \frac{AL_t^\alpha}{L_t} = AL_t^{\alpha-1}$$

Therefore

$$\frac{L_{t+1}}{L_t} = 1 + \gamma (AL_t^{\alpha-1} - c)$$

Shape of the plot (L on the horizontal axis, L_{t+1}/L_t on the vertical):

- **Strictly decreasing** in L , since $\alpha - 1 < 0$.
- As $L \rightarrow 0^+$: $L^{\alpha-1} \rightarrow \infty$, so the growth factor $\rightarrow +\infty$.
- As $L \rightarrow \infty$: $L^{\alpha-1} \rightarrow 0$, so the growth factor $\rightarrow 1 - \gamma c$ (a horizontal asymptote below 1 — population shrinks).
- It crosses the horizontal line at height 1 exactly once, at L^* below.

Draw the horizontal line at 1: the crossing point is the steady state, growth is above it to the left and below it to the right, which shows the dynamics are **globally stable**.

(d) **Neither GDP per capita nor population grows in the long run.**

Set $\frac{L_{t+1}}{L_t} = 1$:

$$\gamma (A(L^*)^{\alpha-1} - c) = 0 \implies A(L^*)^{\alpha-1} = c \implies L^* = \left(\frac{A}{c}\right)^{\frac{1}{1-\alpha}}$$

Stability. For $L < L^*$, $AL^{\alpha-1} > c$, so L grows; for $L > L^*$ it shrinks. Globally stable, converging monotonically. ✓ (Simulation from $L_0 = 500$ with $A = 100$, $c = 2$, $\alpha = 0.5$ converges to $L^* = 2500$ exactly.)

And here is the punchline: at the steady state

$$\frac{C}{L} = A(L^*)^{\alpha-1} = c$$

GDP per capita equals subsistence, always. It cannot be anything else — if it were above, population would grow and push it down; if below, population would fall and push it up. So:

- **GDP per capita: no long-run growth.** It is pinned at c .
- **Population: no long-run growth.** It converges to L^* .

(e) **A one-time increase $A \rightarrow A'$: living standards jump, then fall all the way back. Only population is permanently higher.**

- **Short run.** L is predetermined and cannot jump. So consumption per capita jumps immediately:

$$\frac{C}{L} = A'(L^*)^{\alpha-1} > c$$

✓ In simulation, doubling A from 100 to 200 makes C/L jump from 2.00 to 4.00 on impact. **The generation alive at the moment of the innovation is genuinely better off.**

- **Transition.** Now $C/L > c$, so by (4.5.7) the population grows. As L rises, diminishing returns push $C/L = A'L^{\alpha-1}$ back down.
- **Long run.** The process stops only when C/L is back at c :

$$L_{\text{new}}^* = \left(\frac{A'}{c}\right)^{\frac{1}{1-\alpha}} = L^* \left(\frac{A'}{A}\right)^{\frac{1}{1-\alpha}} > L^*$$

$$\left.\frac{C}{L}\right|_{\text{long run}} = c \quad \text{— exactly as before.}$$

✓ Simulation: A doubles, L^* goes $2500 \rightarrow 10,000$ (a factor of $4 = 2^{1/(1-0.5)}$, as the formula requires), and C/L returns to 2.000000.

This is the Malthusian trap. Technological progress is converted entirely into *more people*, not into better-off people. It is the single most important idea in the exercise, and it explains the central fact of Chapter 3: living standards were roughly flat for millennia despite continuous, if slow, technological improvement. Better ploughs, crop rotation and the horse collar all raised A — and all produced a larger population living at the same subsistence level.

- (f) **With sustained progress $A_{t+1} = A_t(1+g)$: population grows forever, GDP per capita still does not.**

On a balanced path C/L must be constant. Since $C_t/L_t = A_t L_t^{\alpha-1}$, constancy requires $A_t L_t^{\alpha-1}$ constant, i.e. $L_t^{1-\alpha} \propto A_t$, so

$$L_t \propto A_t^{\frac{1}{1-\alpha}} \quad \implies \quad \boxed{\frac{L_{t+1}}{L_t} = (1+g)^{\frac{1}{1-\alpha}}}$$

✓ Simulated growth rate with $g = 0.02$, $\alpha = 0.5$: 0.04040000 , against the theoretical $(1.02)^2 - 1 = 0.04040000$. Exact match.

Population growth is *faster* than technological progress — amplified by $\frac{1}{1-\alpha} > 1$, because it takes a more-than-proportional increase in population to absorb a given productivity gain when returns diminish.

And GDP per capita? Constant — it does not grow. But note it settles *above* subsistence. Setting $L_{t+1}/L_t = (1+g)^{1/(1-\alpha)}$ in (4.5.7):

$$(1+g)^{\frac{1}{1-\alpha}} = 1 + \gamma \left(\frac{C}{L} - c\right) \implies \boxed{\frac{C}{L} = c + \frac{(1+g)^{\frac{1}{1-\alpha}} - 1}{\gamma}}$$

✓ Simulated: 2.808000; formula with $c = 2$, $\gamma = 0.05$: $2 + 0.0404/0.05 = 2.808000$. Exact match.

So sustained technological progress buys a permanently *higher* standard of living than the no-growth case — but a **constant** one. The economy escapes bare subsistence but never achieves growth in income per head. Only breaking the link between living standards and fertility (i.e. $\gamma \rightarrow 0$, the demographic transition) allows modern growth.

(g) **Higher γ means *lower* long-run consumption per capita.**

From the expression just derived,

$$\frac{\partial}{\partial \gamma} \left(\frac{C}{L} \right) = - \frac{(1+g)^{\frac{1}{1-\alpha}} - 1}{\gamma^2} < 0$$

✓ Verified by simulation across a range of γ (with $g = 0.02$, $\alpha = 0.5$, $c = 2$):

γ	Formula	Simulated
0.02	4.020000	4.020000
0.05	2.808000	2.808000
0.10	2.404000	2.404000
0.25	2.161600	2.161600
0.50	2.080800	2.080800

Why. The logic runs backwards from the required population growth rate. Technological progress at rate g *forces* the population to grow at $(1+g)^{1/(1-\alpha)}$ on the balanced path — that rate is determined by technology and α , and γ has no say in it. But equation (4.5.7) says population growth is *produced* by living standards above subsistence. So living standards must sit at whatever level generates the required growth:

- If γ is **high** (population very responsive), only a *small* excess over subsistence is needed to generate that much population growth. Living standards settle close to c .
- If γ is **low** (population sluggish), a *large* excess is needed. Living standards settle well above c .

As $\gamma \rightarrow 0$ the required excess becomes unbounded and the Malthusian mechanism disappears entirely — population no longer responds to income, and technological progress finally shows up as growth in income per head instead of in numbers.

The historical reading. This is exactly the **demographic transition**. What allowed Europe to escape the Malthusian trap after about 1800 was not only faster technological progress but a *collapse in γ* : fertility stopped rising with income and began falling with it. In this model, that is the difference between the world of Chapter 3's flat pre-1800 living standards and the world the Solow model of §4.5 is designed to describe.

```

import numpy as np
alpha, c_sub, gamma = 0.5, 2.0, 0.05

def simulate(A_series, L0, T):
    L, cpc = np.empty(T+1), np.empty(T+1)
    L[0] = L0
    for t in range(T+1):
        cpc[t] = A_series[t] * L[t]**(alpha-1)          # C/L = A L^(alpha-1)
        if t < T:
            L[t+1] = L[t] * (1 + gamma*(cpc[t] - c_sub))
    return L, cpc

# (d) A constant -> L* = (A/c)^(1/(1-alpha)) = 2500, C/L -> c
L, cpc = simulate(np.full(401, 100.0), L0=500.0, T=400)  # L[-1]=2500.0, cpc[-1]=2.0

# (e) one-time A: 100 -> 200 : C/L jumps 2.0 -> 4.0, returns to 2.0; L: 2500 -> 10000
# (f) A grows at g=2% : L grows at (1+g)**(1/(1-alpha))-1 = 0.0404 exactly
#     C/L -> c + ((1+g)**(1/(1-alpha)) - 1)/gamma = 2 + 0.0404/0.05 = 2.808

```

Appendix Cross-reference to the course

The Lista (issued 03/08/2026, due 10/08/2026, 10 points) draws directly on this material:

Q	Pts	Source	See
1	3	Kurlat ch. 1, authored problem	Problem 1.2 (same structure, Brazil/Colombia)
2	2	Kurlat Exercise 2.2	Problem 2.2
3	2	Kurlat Exercise 3.3	Problem 3.3
4	3	Kurlat ch. 4, extended	Problem 4.1 (plus the n halving extension)

Questions 2 and 3 are the book's own exercises assigned *by number*. Question 1 mirrors Problem 1.2 with Brazilian and Colombian data instead of US and Thai. Question 4 is Problem 4.1 with an added part.

Note that Q3 is a data exercise — it requires downloading the Penn World Table and producing three regional scatterplots. Do not leave it to the last day.

Chapter	Aula	Rules file
1–2	1	rules/01_mensuracao_agregados.md
3, 4.1–4.2	2	rules/02_crescimento_solow.md
4.3–4.5	3	rules/03_solow_evidencias.md

Page-level anchors for every section are in Map/registro-livros.md. Kurlat's printed page numbers equal its PDF page numbers (offset 0), so /study-pack can extract any range directly.